

UNIT 1

BASICS OF COMPUTER NETWORKS AND OPERATING SYSTEM:

TOPIC 1 : LAN

1. What does LAN stand for?

- A) Local Access Network
- B) Local Area Network
- C) Large Area Network
- D) Link Access Network

Answer: B) Local Area Network

2. Which of the following devices is used to connect devices in a LAN?

- A) Router
- B) Hub
- C) Switch
- D) Both B and C

Answer: D) Both B and C

3. Which topology is most commonly used in LANs?

- A) Bus
- B) Star
- C) Ring
- D) Mesh

Answer: B) Star

4. What is the primary function of a switch in a LAN?

- A) To connect networks together
- B) To amplify signals
- C) To forward data only to the intended device
- D) To assign IP addresses

Answer: C) To forward data only to the intended device

5. Which protocol is used for communication in a LAN?

- A) HTTP
- B) TCP/IP
- C) SMTP
- D) POP3

Answer: B) TCP/IP

6. What does a DHCP server do in a LAN?

- A) Provides network security
- B) Assigns IP addresses to devices
- C) Encrypts data for secure transmission
- D) Resolves domain names to IP addresses

Answer: B) Assigns IP addresses to devices

7. What is the function of a router in a LAN?

- A) To connect devices within the same network
- B) To connect a LAN to the internet or other networks
- C) To convert analog signals to digital
- D) To connect LANs to wireless devices

Answer: B) To connect a LAN to the internet or other networks

8. Which cable is most commonly used in a wired LAN?

- A) Fiber optic
- B) Ethernet (Cat5e, Cat6)
- C) HDMI
- D) Coaxial

Answer: B) Ethernet (Cat5e, Cat6)

9. What is the maximum transmission distance of a Cat5e Ethernet cable?

- A) 10 meters
- B) 100 meters
- C) 500 meters
- D) 1,000 meters

Answer: B) 100 meters

10. Which device is used to extend the range of a LAN network?

- A) Switch
- B) Repeater
- C) Hub
- D) Router

Answer: B) Repeater

11. What is the typical speed of a Gigabit Ethernet connection in a LAN?

- A) 100 Mbps
- B) 1 Gbps
- C) 10 Gbps
- D) 100 Gbps

Answer: B) 1 Gbps

12. Which of these is a characteristic of a LAN?

- A) It covers a large geographic area
- B) It connects devices over several cities
- C) It typically uses private, owned infrastructure
- D) It only uses wireless communication

Answer: C) It typically uses private, owned infrastructure

13. What does a MAC address identify in a LAN?

- A) The type of network
- B) The physical device on the network
- C) The user of the device
- D) The IP address of the device

Answer: B) The physical device on the network

14. Which of the following is true about Wi-Fi in a LAN?

- A) It connects devices using a physical cable
- B) It is used for long-range network connections
- C) It is a type of wireless LAN technology
- D) It requires a fiber optic cable for communication

Answer: C) It is a type of wireless LAN technology

15. Which of the following protocols is used to assign dynamic IP addresses in a LAN?

- A) HTTP
- B) DHCP
- C) DNS
- D) FTP

Answer: B) DHCP

16. Which layer of the OSI model is responsible for transmitting raw data bits over a physical medium?

- A) Application Layer
- B) Transport Layer
- C) Network Layer
- D) Physical Layer

Answer: D) Physical Layer

17. What is the function of a hub in a LAN?

- A) To assign IP addresses
- B) To broadcast data to all connected devices
- C) To route data between different networks
- D) To filter and forward data

Answer: B) To broadcast data to all connected devices

18. What is the purpose of using a firewall in a LAN?

- A) To assign IP addresses
- B) To block unauthorized access to the network
- C) To route data packets between LANs
- D) To provide wireless access

Answer: B) To block unauthorized access to the network

19. Which of the following is an example of a wireless LAN (WLAN) standard?

- A) 802.3
- B) 802.11b
- C) 802.15
- D) 802.5

Answer: B) 802.11b

20. What is the maximum data transfer rate of Fast Ethernet (100Base-T)?

- A) 10 Mbps
- B) 100 Mbps
- C) 1 Gbps
- D) 10 Gbps

Answer: B) 100 Mbps

21. Which device is used to filter network traffic based on IP addresses in a LAN?

- A) Hub
- B) Switch
- C) Firewall
- D) Repeater

Answer: C) Firewall

22. What is the function of a DNS server in a LAN?

- A) To assign IP addresses
- B) To resolve domain names into IP addresses
- C) To route traffic between networks
- D) To block spam emails

Answer: B) To resolve domain names into IP addresses

23. What is the function of a Network Interface Card (NIC)?

- A) To assign IP addresses
- B) To connect devices to a network
- C) To manage network traffic
- D) To filter network data

Answer: B) To connect devices to a network

24. Which technology is used in LANs to share files and printers?

- A) FTP
- B) TCP/IP
- C) SMB
- D) SSH

Answer: C) SMB

25. What is the maximum distance of a wireless LAN (Wi-Fi) connection in a typical office environment?

- A) 50 meters
- B) 100 meters
- C) 500 meters
- D) 1,000 meters

Answer: A) 50 meters

26. What is the role of a bridge in a LAN?

- A) To connect two or more LAN segments
- B) To filter data based on MAC addresses
- C) To connect a LAN to a wide-area network
- D) To amplify signals over long distances

Answer: A) To connect two or more LAN segments

27. Which of the following is true about Ethernet technology in LANs?

- A) It uses radio waves to transmit data
- B) It is a wired networking technology

- C) It requires a router to operate
- D) It can only support wireless devices

Answer: B) It is a wired networking technology

28. Which of the following is a commonly used IP addressing scheme in a LAN?

- A) Private IP addresses
- B) Dynamic IP addresses
- C) Public IP addresses
- D) IPv4 addresses only

Answer: A) Private IP addresses

29. Which of the following is NOT a feature of a switch in a LAN?

- A) It forwards data based on MAC addresses
- B) It increases the number of collision domains
- C) It operates at Layer 3 of the OSI model
- D) It helps improve network efficiency

Answer: C) It operates at Layer 3 of the OSI model

30. Which of the following is NOT a common type of wireless network security?

- A) WPA2
- B) WEP
- C) DNS
- D) WPA

Answer: C) DNS

31. Which of these devices is used to combine multiple Ethernet connections into one in a LAN?

- A) Switch
- B) Hub
- C) Bridge
- D) Router

Answer: A) Switch

32. What type of address is used by switches to forward Ethernet frames?

- A) IP Address
- B) MAC Address
- C) Port Address
- D) DNS Address

Answer: B) MAC Address

33. Which of the following cables can support the highest bandwidth in a LAN?

- A) Coaxial
- B) Fiber optic
- C) Ethernet (Cat 5e)
- D) Ethernet (Cat 6)

Answer: B) Fiber optic

34. What is the main purpose of using VLANs in a LAN?

- A) To increase the bandwidth of the LAN
- B) To secure and segment network traffic
- C) To replace physical switches
- D) To reduce the number of devices connected to a LAN

Answer: B) To secure and segment network traffic

35. What is the typical speed of a 10G Ethernet connection in a LAN?

- A) 100 Mbps
- B) 1 Gbps
- C) 10 Gbps
- D) 100 Gbps

Answer: C) 10 Gbps

36. What is the primary difference between a hub and a switch in a LAN?

- A) A hub routes data while a switch does not
- B) A hub broadcasts data to all devices, while a switch sends data only to the intended device
- C) A hub works at Layer 3 of the OSI model, while a switch works at Layer 2
- D) A switch cannot be used in a LAN, but a hub can

Answer: B) A hub broadcasts data to all devices, while a switch sends data only to the intended device

37. Which of the following is NOT a common type of network topology in a LAN?

- A) Star
- B) Ring
- C) Tree
- D) Cloud

Answer: D) Cloud

38. Which of the following is the default port number for HTTP in a LAN?

- A) 25
- B) 80
- C) 443
- D) 21

Answer: B) 80

39. What is the primary function of a gateway in a LAN?

- A) To connect the LAN to the internet
- B) To route traffic within the LAN
- C) To provide wireless access
- D) To assign IP addresses

Answer: A) To connect the LAN to the internet

40. Which of the following technologies is used for short-range wireless communication in LANs?

- A) Wi-Fi
- B) Bluetooth
- C) Zigbee
- D) 4G LTE

Answer: B) Bluetooth

41. What type of IP address range is used by private LANs?

- A) 0.0.0.0 to 255.255.255.255
- B) 10.0.0.0 to 10.255.255.255
- C) 172.16.0.0 to 172.31.255.255
- D) 192.168.0.0 to 192.168.255.255

Answer: B) 10.0.0.0 to 10.255.255.255

42. What is the main role of an access point (AP) in a wireless LAN?

- A) To provide wired internet access
- B) To connect multiple LANs together
- C) To extend the range of a wireless network
- D) To convert wired traffic into wireless traffic

Answer: D) To convert wired traffic into wireless traffic

43. Which of the following is a characteristic of a Layer 3 switch?

- A) It routes data between networks
- B) It works only with wired connections
- C) It filters data based on MAC addresses
- D) It cannot assign IP addresses

Answer: A) It routes data between networks

44. Which of these protocols is used for secure communication in a LAN?

- A) SMTP
- B) SSH
- C) HTTP
- D) IMAP

Answer: B) SSH

45. Which of the following standards is used for Gigabit Ethernet in a LAN?

- A) 802.3u
- B) 802.3z
- C) 802.3ab
- D) 802.11g

Answer: C) 802.3ab

46. What is the maximum data rate supported by 802.11n Wi-Fi in a LAN?

- A) 54 Mbps
- B) 100 Mbps
- C) 600 Mbps
- D) 1 Gbps

Answer: C) 600 Mbps

47. What does an IP address identify in a LAN?

- A) The physical device
- B) The port number
- C) The network interface of a device
- D) The protocol being used

Answer: C) The network interface of a device

48. Which layer of the OSI model is responsible for reliable data transfer in a LAN?

- A) Application Layer
- B) Transport Layer
- C) Network Layer
- D) Data Link Layer

Answer: B) Transport Layer

49. Which of the following is true about the data link layer in a LAN?

- A) It routes data across networks
- B) It is responsible for error checking and frame synchronization
- C) It defines the path between two devices
- D) It encrypts data for security

Answer: B) It is responsible for error checking and frame synchronization

50. What is the main advantage of fiber optic cables over copper cables in LANs?

- A) Lower cost
- B) Higher bandwidth and longer range
- C) Easier to install
- D) Better resistance to interference

Answer: B) Higher bandwidth and longer range

TOPIC 2 : WAN

1. Which of the following is a characteristic of a WAN?

- a) Limited to a small geographic area
- b) Covers a large geographic area
- c) Uses only wireless communication
- d) Connects only a few devices

Answer: b) Covers a large geographic area

2. What does WAN stand for?

- a) Wireless Area Network
- b) Wide Area Network
- c) Web Area Network
- d) Work Area Network

Answer: b) Wide Area Network

3. Which device is commonly used to connect devices in a WAN?

- a) Switch
- b) Hub
- c) Router
- d) Bridge

Answer: c) Router

4. Which of the following technologies is commonly used for WAN connections?

- a) Bluetooth

- b) Ethernet
- c) Fiber Optic
- d) USB

Answer: c) Fiber Optic

5. Which of the following is an example of a WAN?

- a) Home Wi-Fi
- b) LAN in a school
- c) The internet
- d) Bluetooth network

Answer: c) The internet

6. Which of the following is true about WANs?

- a) WANs are faster than LANs
- b) WANs use public telecommunication lines
- c) WANs are not scalable
- d) WANs operate only in a single building

Answer: b) WANs use public telecommunication lines

7. Which of the following is a type of WAN connection?

- a) VPN
- b) LAN
- c) WLAN
- d) MAN

Answer: a) VPN

8. Which of these is a disadvantage of WANs?

- a) Lower cost
- b) Complex setup
- c) Easier to secure
- d) Limited scalability

Answer: b) Complex setup

9. What is a common use of a WAN?

- a) To connect devices in a small area like a home or office
- b) To connect distant networks in different cities or countries
- c) To transmit data wirelessly over short distances

- d) To connect only a few computers in a local area

Answer: b) To connect distant networks in different cities or countries

10. Which of the following is not a characteristic of WANs?

- a) They are geographically dispersed
- b) They require high-bandwidth communication
- c) They only connect computers within a single organization
- d) They can connect different types of networks

Answer: c) They only connect computers within a single organization

11. What is the primary purpose of a router in a WAN?

- a) Connect computers to the internet
- b) Route data packets between different networks
- c) Amplify signals over long distances
- d) Manage network traffic within a local area

Answer: b) Route data packets between different networks

12. Which of the following technologies is typically used to establish a WAN link?

- a) Wi-Fi
- b) Ethernet
- c) DSL
- d) USB

Answer: c) DSL

13. What is a major advantage of WANs?

- a) Lower cost of implementation
- b) Ability to connect multiple locations globally
- c) High-speed data transfer within a single location
- d) Limited to a small area

Answer: b) Ability to connect multiple locations globally

14. Which of the following is a widely used WAN protocol?

- a) HTTP
- b) SMTP
- c) PPP
- d) FTP

Answer: c) PPP

15. Which technology allows remote users to securely access a WAN?

- a) FTP
- b) VPN
- c) HTTP
- d) DHCP

Answer: b) VPN

16. Which of the following can be used to increase the reliability of a WAN?

- a) Bandwidth throttling
- b) Redundant communication links
- c) Limited routing protocols
- d) Restricting external access

Answer: b) Redundant communication links

17. In which scenario would a WAN be most useful?

- a) Connecting computers within a single office building
- b) Connecting multiple branch offices of a company
- c) Connecting devices in a home network
- d) Connecting wireless devices in a small area

Answer: b) Connecting multiple branch offices of a company

18. Which of the following is true about WAN bandwidth?

- a) WAN bandwidth is always fixed and cannot be changed
- b) WAN bandwidth can be shared among multiple users
- c) WANs do not require any bandwidth
- d) WAN bandwidth is independent of traffic demand

Answer: b) WAN bandwidth can be shared among multiple users

19. Which of the following is NOT typically a WAN connection medium?

- a) Satellite link
- b) Fiber optic cable
- c) Coaxial cable
- d) Bluetooth

Answer: d) Bluetooth

20. Which of these is a benefit of using a WAN for communication?

- a) Limited coverage area
- b) Lower installation cost
- c) Enhanced ability to share data across regions
- d) Improved security

Answer: c) Enhanced ability to share data across regions

21. Which of the following is a key feature of WANs?

- a) They are generally used for connecting networks within a building
- b) They use low-speed data transmission
- c) They can connect remote locations over large distances
- d) They are always wireless

Answer: c) They can connect remote locations over large distances

22. What is the main difference between a LAN and a WAN?

- a) LANs cover larger geographical areas
- b) LANs are faster than WANs
- c) WANs use wireless technology, while LANs use wired technology
- d) LANs cover smaller geographical areas compared to WANs

Answer: d) LANs cover smaller geographical areas compared to WANs

23. Which of the following is used to create a virtual WAN over a public network?

- a) DSL
- b) VPN
- c) Bluetooth
- d) Ethernet

Answer: b) VPN

24. Which protocol is used to connect devices in a WAN over a telephone line?

- a) HTTP
- b) POP3
- c) PPP
- d) SMTP

Answer: c) PPP

25. Which of these is an example of a WAN service provider?

- a) AT&T
- b) Wi-Fi hotspot
- c) Bluetooth network
- d) Ethernet cable company

Answer: a) AT&T

26. What is the most common use of WAN in a corporate environment?

- a) Sharing files within a local office
- b) Connecting global branch offices
- c) Transmitting data to nearby devices
- d) Connecting personal devices in a home

Answer: b) Connecting global branch offices

27. Which of the following is a major WAN disadvantage?

- a) Long setup time
- b) High cost
- c) Limited reach
- d) Slow data transmission

Answer: b) High cost

28. Which of the following WAN technologies provides internet access to remote locations?

- a) DSL
- b) Wi-Fi
- c) Bluetooth
- d) VPN

Answer: a) DSL

29. Which of the following is the primary function of a WAN?

- a) To allow users to access the internet
- b) To interconnect computers within a building
- c) To link computers in a local area network (LAN)
- d) To connect devices across large distances or regions

Answer: d) To connect devices across large distances or regions

30. Which of these is true about WAN routers?

- a) WAN routers are only used in small offices
- b) WAN routers manage data traffic between different networks
- c) WAN routers only work with fiber optic cables
- d) WAN routers cannot handle high bandwidth

Answer: b) WAN routers manage data traffic between different networks

31. Which WAN technology uses a dedicated leased line between two points?

- a) Frame Relay
- b) ATM
- c) MPLS
- d) Private Line

Answer: d) Private Line

32. Which type of WAN technology is considered reliable and flexible for connecting branch offices?

- a) DSL
- b) VPN
- c) Ethernet
- d) MPLS

Answer: d) MPLS

33. Which of the following is NOT used for WAN connections?

- a) Microwave transmission
- b) Satellite
- c) Ethernet
- d) Coaxial cable

Answer: c) Ethernet

34. What does MPLS stand for in WAN technology?

- a) Managed Private Link Service
- b) Multi-Protocol Label Switching
- c) Multi-Point Link Service
- d) Multi-Purpose Labeling System

Answer: b) Multi-Protocol Label Switching

35. Which of the following is the most common technology for internet-based WAN?

- a) Wi-Fi
- b) Satellite internet

- c) Ethernet
- d) Fiber optics

Answer: b) Satellite internet

36. Which of the following WAN technologies supports the highest speeds?

- a) ISDN
- b) DSL
- c) Fiber Optic
- d) Wi-Fi

Answer: c) Fiber Optic

37. Which of the following is a disadvantage of satellite-based WAN links?

- a) High latency
- b) Low cost
- c) High speed
- d) Unlimited data transfer

Answer: a) High latency

38. Which of the following is true about WANs using MPLS?

- a) They are always cheaper than traditional WANs
- b) They are limited to voice communication only
- c) They provide a faster and more efficient data transmission
- d) They do not allow data encryption

Answer: c) They provide a faster and more efficient data transmission

39. What does the acronym WAN stand for in networking?

- a) Wireless Area Network
- b) World Area Network
- c) Wide Area Network
- d) Web Access Network

Answer: c) Wide Area Network

40. What is a key characteristic of a WAN compared to a LAN?

- a) It only works within a single building
- b) It uses wireless communication only
- c) It can cover large distances, such as across cities or countries
- d) It connects fewer devices

Answer: c) It can cover large distances, such as across cities or countries

41. Which of the following WAN technologies offers high-speed internet connections for remote locations?

- a) DSL
- b) VPN
- c) MPLS
- d) Ethernet

Answer: a) DSL

42. Which of the following is NOT a characteristic of a WAN?

- a) It covers large geographic areas
- b) It connects multiple local area networks
- c) It uses high-speed connections between devices
- d) It is restricted to one office building

Answer: d) It is restricted to one office building

43. Which of the following WAN technologies is commonly used for businesses that need secure connections to remote locations?

- a) Frame Relay
- b) VPN
- c) ISDN
- d) DSL

Answer: b) VPN

44. Which of these is used to connect local area networks over a large geographic area in a WAN?

- a) Fiber optics
- b) Ethernet cables
- c) Satellite
- d) Wi-Fi

Answer: a) Fiber optics

45. Which of the following is an advantage of using WANs?

- a) They cover a limited geographic area
- b) They allow remote access to company networks
- c) They do not require security measures
- d) They can only be used in homes

Answer: b) They allow remote access to company networks

46. Which of these WAN types is typically used by large-scale organizations for connecting branch offices?

- a) Ethernet WAN
- b) VPN WAN
- c) MPLS WAN
- d) DSL WAN

Answer: c) MPLS WAN

47. Which of the following is an example of a public WAN?

- a) The internet
- b) A private office network
- c) A company LAN
- d) A closed circuit network

Answer: a) The internet

48. Which type of device connects a local area network to a WAN?

- a) Router
- b) Switch
- c) Hub
- d) Repeater

Answer: a) Router

49. Which of the following is NOT a primary function of WAN?

- a) Providing internet access
- b) Connecting remote offices
- c) Connecting personal devices in a small space
- d) Connecting geographically separated networks

Answer: c) Connecting personal devices in a small space

50. Which of the following WAN technologies is used to connect networks across different cities or countries?

- a) Satellite
- b) LAN
- c) Wi-Fi
- d) Ethernet

Answer: a) Satellite

TOPIC 3 : WIRELESS NETWORKS

1. What does Wi-Fi stand for?

- **A. Wireless Fidelity**
- B. Wide Frequency
- C. Wireless Frequency
- D. None of the above

Answer: A. Wireless Fidelity

2. Which IEEE standard is used for wireless networking?

- A. 802.3
- **B. 802.11**
- C. 802.15
- D. 802.16

Answer: B. 802.11

3. What frequency bands are commonly used in Wi-Fi?

- A. 1.5 GHz and 3.2 GHz
- **B. 2.4 GHz and 5 GHz**
- C. 4.9 GHz and 6 GHz
- D. 2 GHz and 3 GHz

Answer: B. 2.4 GHz and 5 GHz

4. What does SSID stand for in Wi-Fi networks?

- A. Signal Strength Identifier
- **B. Service Set Identifier**
- C. Secure Signal Identifier
- D. Shared Signal Index

Answer: B. Service Set Identifier

5. What is the purpose of a wireless access point?

- A. To charge wireless devices
- **B. To connect wireless devices to a wired network**
- C. To manage IP addresses
- D. To boost internet speed

Answer: B. To connect wireless devices to a wired network

6. Which wireless security protocol is the most secure?

- A. WEP
- B. WPA
- C. WPA2

- **D. WPA3**
Answer: D. WPA3
-

7. What does WPA stand for?

- A. Wireless Protection Algorithm
 - **B. Wi-Fi Protected Access**
 - C. Wireless Protected Authentication
 - D. Wi-Fi Protocol Authentication
- Answer: B. Wi-Fi Protected Access**
-

8. What is the typical range of a Wi-Fi network indoors?

- A. 5-10 meters
 - **B. 30-50 meters**
 - C. 100-150 meters
 - D. 200-300 meters
- Answer: B. 30-50 meters**
-

9. What is the function of a wireless router?

- A. To provide a firewall
 - B. To connect devices using Ethernet
 - **C. To distribute internet wirelessly**
 - D. To encrypt data packets
- Answer: C. To distribute internet wirelessly**
-

10. Which device is used to extend the range of a wireless network?

- A. Router
 - B. Switch
 - **C. Wi-Fi Repeater**
 - D. Hub
- Answer: C. Wi-Fi Repeater**
-

11. Which technology enables devices to communicate directly without a router?

- A. Bluetooth
 - **B. Wi-Fi Direct**
 - C. Zigbee
 - D. NFC
- Answer: B. Wi-Fi Direct**
-

12. What does the term "Hotspot" refer to in networking?

- A. A physical location for routers

- B. A device for data transfer
 - **C. A public wireless access point**
 - D. A type of antenna
- Answer: C. A public wireless access point**
-

13. What type of network topology does Wi-Fi use?

- A. Bus
 - **B. Star**
 - C. Ring
 - D. Mesh
- Answer: B. Star**
-

14. Which wireless standard supports the fastest data transfer rates?

- A. 802.11g
 - B. 802.11n
 - C. 802.11ac
 - **D. 802.11ax**
- Answer: D. 802.11ax**
-

15. What is the full form of LAN?

- A. Local Area Network
 - **B. Local Access Network**
 - C. Linked Area Network
 - D. Limited Access Network
- Answer: A. Local Area Network**
-

16. Which protocol is used for secure data transmission in wireless networks?

- A. FTP
 - B. HTTP
 - **C. HTTPS**
 - D. DNS
- Answer: C. HTTPS**
-

17. What does Bluetooth use for communication?

- A. Infrared
 - **B. Radio Waves**
 - C. Wi-Fi signals
 - D. Microwaves
- Answer: B. Radio Waves**
-

18. What is the maximum theoretical speed of the 802.11ac standard?

- A. 600 Mbps
- **B. 1.3 Gbps**
- C. 450 Mbps
- D. 100 Mbps

Answer: B. 1.3 Gbps

19. Which frequency band provides better range but lower speed?

- **A. 2.4 GHz**
- B. 5 GHz
- C. 6 GHz
- D. 1 GHz

Answer: A. 2.4 GHz

20. What is an ad-hoc wireless network?

- A. A centralized network
- **B. A peer-to-peer network**
- C. A satellite network
- D. A 5G network

Answer: B. A peer-to-peer network

21. Which layer of the OSI model handles wireless communication?

- A. Application Layer
- **B. Physical Layer**
- C. Network Layer
- D. Data Link Layer

Answer: B. Physical Layer

22. What is the term for interference caused by multiple overlapping wireless signals?

- A. Noise
- **B. Co-channel Interference**
- C. Signal Jamming
- D. Cross-talk

Answer: B. Co-channel Interference

23. What does NFC stand for in wireless communication?

- A. Near Field Connection
- **B. Near Field Communication**
- C. Network Frequency Channel
- D. None of the above

Answer: B. Near Field Communication

24. What is the primary purpose of a firewall in a wireless network?

- A. To manage bandwidth
- B. To improve speed
- **C. To enhance security**
- D. To expand coverage

Answer: C. To enhance security

25. What does the acronym IoT stand for?

- A. Internet of Traffic
- B. Input Output Technology
- **C. Internet of Things**
- D. Intelligent Online Transfer

Answer: C. Internet of Things

26. What is the purpose of encryption in wireless networks?

- A. To improve speed
- B. To manage IP addresses
- **C. To protect data from unauthorized access**
- D. To reduce network traffic

Answer: C. To protect data from unauthorized access

27. Which wireless technology is used for very short-range communication, such as contactless payments?

- A. Wi-Fi
- **B. NFC**
- C. Zigbee
- D. Bluetooth

Answer: B. NFC

28. What does "bandwidth" refer to in networking?

- **A. The amount of data that can be transmitted in a given time**
- B. The range of frequencies available for use
- C. The strength of a wireless signal
- D. The maximum range of a network

Answer: A. The amount of data that can be transmitted in a given time

29. Which of these is NOT a wireless networking standard?

- A. 802.11b
- B. 802.11ac
- **C. 802.3**
- D. 802.11n

Answer: C. 802.3

30. What is the primary advantage of the 5 GHz band over 2.4 GHz?

- A. Better range
- **B. Higher speed and less interference**
- C. Compatibility with older devices
- D. Stronger signal strength

Answer: B. Higher speed and less interference

31. What is a "MAC address"?

- A. A device's IP address
 - **B. A unique identifier for network devices**
 - C. A network security protocol
 - D. A type of encryption
- Answer: B. A unique identifier for network devices**
-

32. What is a wireless mesh network?

- **A. A network where devices connect through multiple nodes**
 - B. A network with centralized access points
 - C. A temporary peer-to-peer connection
 - D. A high-speed LAN network
- Answer: A. A network where devices connect through multiple nodes**
-

33. What does the term "ping" measure in networking?

- A. Signal strength
 - **B. Network latency**
 - C. Data transfer rate
 - D. Encryption level
- Answer: B. Network latency**
-

34. What is the maximum theoretical speed of Wi-Fi 6 (802.11ax)?

- A. 1.3 Gbps
 - B. 4.8 Gbps
 - **C. 9.6 Gbps**
 - D. 6.9 Gbps
- Answer: C. 9.6 Gbps**
-

35. What is the term for a situation where multiple wireless networks interfere with each other?

- **A. Spectrum Congestion**
- B. Signal Attenuation
- C. Noise

- D. Signal Reflection
- Answer: A. Spectrum Congestion**
-

36. What is the role of DHCP in wireless networks?

- A. Encrypts network communication
 - **B. Assigns IP addresses automatically**
 - C. Monitors network security
 - D. Controls data transfer rates
- Answer: B. Assigns IP addresses automatically**
-

37. What is the full form of WEP?

- A. Wireless Encrypted Protocol
 - **B. Wired Equivalent Privacy**
 - C. Wireless Enhanced Protection
 - D. Wi-Fi Encryption Protocol
- Answer: B. Wired Equivalent Privacy**
-

38. Which technology is commonly used for IoT devices?

- A. Wi-Fi
 - B. Bluetooth
 - **C. Zigbee**
 - D. LTE
- Answer: C. Zigbee**
-

39. What is the purpose of the 802.1X standard in wireless networks?

- A. Improve range
 - **B. Provide authentication and access control**
 - C. Enhance speed
 - D. Increase signal strength
- Answer: B. Provide authentication and access control**
-

40. What is the typical range of Bluetooth communication?

- A. 1-5 meters
 - **B. 10 meters**
 - C. 50 meters
 - D. 100 meters
- Answer: B. 10 meters**
-

41. What does MIMO stand for in wireless networks?

- **A. Multiple Input Multiple Output**

- B. Multi-Interface Multi-Output
- C. Multi Integrated Modulation Output
- D. Modular Input Mixed Output

Answer: A. Multiple Input Multiple Output

42. What is the primary function of a wireless adapter?

- A. Increase network speed
- **B. Connect devices to a wireless network**
- C. Extend network range
- D. Manage network protocols

Answer: B. Connect devices to a wireless network

43. What is "packet loss" in networking?

- A. Data encryption failure
- B. Loss of internet connection
- **C. Failure of data packets to reach their destination**
- D. Corruption of transmitted data

Answer: C. Failure of data packets to reach their destination

44. Which of the following is NOT a wireless encryption protocol?

- A. WPA
- **B. IPsec**
- C. WEP
- D. WPA2

Answer: B. IPsec

45. What is beamforming in wireless communication?

- A. Reducing signal interference
- **B. Directing the signal toward specific devices**
- C. Increasing network bandwidth
- D. Splitting signals into multiple streams

Answer: B. Directing the signal toward specific devices

46. What does LTE stand for?

- **A. Long-Term Evolution**
- B. Local Transmission Extension
- C. Low Transmission Efficiency
- D. Limited Transmission Exchange

Answer: A. Long-Term Evolution

47. What is the primary difference between Wi-Fi and Bluetooth?

- **A. Wi-Fi is designed for high-speed data transfer over long distances, while Bluetooth is for short-range communication.**
- B. Bluetooth is faster than Wi-Fi.
- C. Wi-Fi uses infrared signals.
- D. Bluetooth is more secure than Wi-Fi.

Answer: A. Wi-Fi is designed for high-speed data transfer over long distances, while Bluetooth is for short-range communication.

48. What is a rogue access point?

- A. A secure access point
- **B. An unauthorized wireless access point**
- C. A high-speed network router
- D. A repeater used for Wi-Fi

Answer: B. An unauthorized wireless access point

49. What is signal attenuation?

- **A. Weakening of signal strength over distance**
- B. Strengthening of signals
- C. Loss of data during transmission
- D. Signal interference

Answer: A. Weakening of signal strength over distance

50. What is the purpose of a VPN in wireless networking?

- A. Extend network range
- **B. Securely connect to a network remotely**
- C. Improve data transfer speeds
- D. Provide additional bandwidth

Answer: B. Securely connect to a network remotely

TOPIC 4 :WLAN

1. What does WLAN stand for?

- A. Wide Local Area Network
- **B. Wireless Local Area Network**
- C. Wide Linear Access Network
- D. Wireless Linear Access Network

Answer: B. Wireless Local Area Network

2. Which IEEE standard defines WLAN?

- A. 802.15
- **B. 802.11**
- C. 802.16
- D. 802.3

Answer: B. 802.11

3. What is the primary purpose of a WLAN?

- A. To connect devices using cables
- **B. To connect devices wirelessly within a local area**
- C. To connect networks over long distances
- D. To connect servers directly to storage devices

Answer: B. To connect devices wirelessly within a local area

4. What frequency bands are commonly used in WLANs?

- A. 1.5 GHz and 2 GHz
 - **B. 2.4 GHz and 5 GHz**
 - C. 3 GHz and 4 GHz
 - D. 5 GHz and 6 GHz
- Answer: B. 2.4 GHz and 5 GHz**
-

5. What is the role of an access point in a WLAN?

- A. To encrypt data
 - **B. To provide wireless connectivity to devices**
 - C. To manage bandwidth
 - D. To distribute IP addresses
- Answer: B. To provide wireless connectivity to devices**
-

6. What does SSID stand for in WLANs?

- A. Secure System Identification
 - B. Signal Strength Identification
 - **C. Service Set Identifier**
 - D. Shared System Identifier
- Answer: C. Service Set Identifier**
-

7. Which of these is NOT a benefit of WLANs?

- A. Mobility
 - B. Easy installation
 - **C. High latency**
 - D. Scalability
- Answer: C. High latency**
-

8. Which protocol is used to secure WLAN communication?

- A. FTP
- B. HTTP
- **C. WPA2/WPA3**

- D. SMTP
- Answer: C. WPA2/WPA3**
-

9. What does a wireless repeater do in a WLAN?

- A. Encrypts data packets
 - **B. Extends the range of the WLAN**
 - C. Assigns IP addresses
 - D. Boosts network speed
- Answer: B. Extends the range of the WLAN**
-

10. What type of network topology is commonly used in WLANs?

- A. Bus
 - B. Mesh
 - **C. Star**
 - D. Ring
- Answer: C. Star**
-

11. What does a WLAN controller do?

- A. Improves signal strength
 - B. Encrypts all communication
 - **C. Manages multiple access points in a network**
 - D. Boosts network speed
- Answer: C. Manages multiple access points in a network**
-

12. What is the maximum theoretical speed of Wi-Fi 6 in WLANs?

- A. 3.2 Gbps
 - B. 6.9 Gbps
 - **C. 9.6 Gbps**
 - D. 12 Gbps
- Answer: C. 9.6 Gbps**
-

13. What does WEP stand for in WLAN security?

- **A. Wired Equivalent Privacy**
 - B. Wireless Encrypted Protection
 - C. Wi-Fi Enhanced Protocol
 - D. Wireless Encryption Protocol
- Answer: A. Wired Equivalent Privacy**
-

14. What is the purpose of a MAC address in WLANs?

- A. To encrypt data

- **B. To uniquely identify a device**
 - C. To determine signal strength
 - D. To assign IP addresses
- Answer: B. To uniquely identify a device**
-

15. What causes interference in WLANs?

- **A. Overlapping signals from other networks**
 - B. Strong encryption protocols
 - C. Using a 5 GHz frequency
 - D. Using multiple access points
- Answer: A. Overlapping signals from other networks**
-

16. What does the term "hotspot" refer to in WLAN?

- A. A secure wireless network
 - B. A mobile data connection
 - **C. A location providing wireless internet access**
 - D. A type of network cable
- Answer: C. A location providing wireless internet access**
-

17. Which WLAN standard introduced MIMO technology?

- A. 802.11g
 - B. 802.11a
 - **C. 802.11n**
 - D. 802.11b
- Answer: C. 802.11n**
-

18. What is the function of a DHCP server in a WLAN?

- **A. Automatically assigns IP addresses**
 - B. Encrypts wireless data
 - C. Increases signal strength
 - D. Manages bandwidth
- Answer: A. Automatically assigns IP addresses**
-

19. What is an ad-hoc WLAN?

- A. A centralized network with access points
 - **B. A temporary peer-to-peer wireless network**
 - C. A high-speed 5G network
 - D. A network with multiple access points
- Answer: B. A temporary peer-to-peer wireless network**
-

20. What does WPA stand for in WLAN security?

- A. Wireless Protection Algorithm
 - **B. Wi-Fi Protected Access**
 - C. Wireless Protected Authentication
 - D. Wireless Protocol Access
- Answer: B. Wi-Fi Protected Access**
-

21. Which WLAN frequency band offers better range?

- **A. 2.4 GHz**
 - B. 5 GHz
 - C. 6 GHz
 - D. 1 GHz
- Answer: A. 2.4 GHz**
-

22. What is the role of beamforming in WLANs?

- A. Splitting data streams
 - B. Increasing data rates
 - **C. Directing the signal towards specific devices**
 - D. Expanding network range
- Answer: C. Directing the signal towards specific devices**
-

23. What does the term "throughput" refer to in WLANs?

- A. Signal strength
 - **B. The actual data transfer rate**
 - C. The number of devices connected
 - D. The range of the network
- Answer: B. The actual data transfer rate**
-

24. What is a rogue access point in a WLAN?

- A. A secure network device
 - B. A backup access point
 - **C. An unauthorized access point**
 - D. A repeater used for coverage
- Answer: C. An unauthorized access point**
-

25. Which encryption standard replaced WEP in WLANs?

- A. HTTPS
 - **B. WPA**
 - C. FTP
 - D. Telnet
- Answer: B. WPA**
-

26. What is the maximum range of a typical indoor WLAN?

- A. 10-20 meters
- **B. 30-50 meters**
- C. 100-200 meters
- D. 300-500 meters

Answer: B. 30-50 meters

27. What does "Wi-Fi" stand for?

- A. Wireless Fidelity
- B. Wireless Internet
- **C. It does not officially stand for anything**
- D. Wireless Interface

Answer: C. It does not officially stand for anything

28. What is the function of an SSID in WLANs?

- A. Assign IP addresses
- B. Encrypt network traffic
- **C. Identify the wireless network**
- D. Extend the network range

Answer: C. Identify the wireless network

29. Which WLAN standard operates exclusively in the 5 GHz band?

- A. 802.11g
- B. 802.11b
- **C. 802.11a**
- D. 802.11n

Answer: C. 802.11a

30. Which WLAN technology provides the best security?

- A. WEP
- **B. WPA3**
- C. WPA
- D. Open Network

Answer: B. WPA3

31. What is a wireless bridge used for in WLANs?

- A. Extending the signal range
- B. Encrypting wireless traffic
- **C. Connecting two networks wirelessly**
- D. Increasing network speed

Answer: C. Connecting two networks wirelessly

32. Which device connects a WLAN to a wired LAN?

- **A. Wireless router**
- B. Wireless repeater
- C. Wireless adapter
- D. Wireless extender

Answer: A. Wireless router

33. What does 802.11ax prioritize compared to earlier standards?

- A. Higher range
- B. Simplified encryption
- **C. Improved efficiency in dense environments**
- D. Compatibility with older devices

Answer: C. Improved efficiency in dense environments

34. What is the main disadvantage of using the 5 GHz band in WLANs?

- A. Higher latency
- **B. Shorter range**
- C. Incompatibility with older devices
- D. Slower data rates

Answer: B. Shorter range

35. What does a wireless adapter do?

- A. Extends wireless signal range
- **B. Enables devices to connect to a WLAN**
- C. Assigns IP addresses
- D. Acts as a wireless repeater

Answer: B. Enables devices to connect to a WLAN

36. What is a Wi-Fi hotspot?

- A. A secure private network
- **B. A location offering wireless internet access**
- C. A wireless signal repeater
- D. A mobile device sharing data

Answer: B. A location offering wireless internet access

37. What is the function of a WLAN gateway?

- A. Encrypting data packets
- **B. Connecting a WLAN to external networks**
- C. Managing device authentication

- D. Improving network throughput
- Answer: B. Connecting a WLAN to external networks**
-

38. What does "handshaking" mean in WLANs?

- **A. Establishing a connection between devices**
 - B. Disconnecting devices from the network
 - C. Encrypting data packets
 - D. Extending signal range
- Answer: A. Establishing a connection between devices**
-

39. What does QoS stand for in WLANs?

- A. Quality of Security
 - **B. Quality of Service**
 - C. Quick Operational Speed
 - D. Queue Optimization System
- Answer: B. Quality of Service**
-

40. What is a captive portal in WLANs?

- A. A security protocol
 - **B. A web page requiring user authentication**
 - C. A method of encryption
 - D. A device that connects to a router
- Answer: B. A web page requiring user authentication**
-

41. What does OFDM stand for in WLANs?

- A. Orthogonal Frequency Division Mechanism
 - **B. Orthogonal Frequency Division Multiplexing**
 - C. Optimized Frequency Data Modulation
 - D. Offset Frequency Data Management
- Answer: B. Orthogonal Frequency Division Multiplexing**
-

42. What is the purpose of roaming in WLANs?

- A. Encrypting data packets
 - B. Extending the network range
 - **C. Seamlessly switching between access points**
 - D. Boosting signal strength
- Answer: C. Seamlessly switching between access points**
-

43. What is a wireless SSID broadcast?

- A. A security feature for networks

- **B. The transmission of the network name**
 - C. A method of extending range
 - D. An encryption protocol
- Answer: B. The transmission of the network name**
-

44. What is the function of a wireless LAN controller?

- A. Encrypting all network data
 - **B. Managing multiple access points**
 - C. Assigning IP addresses
 - D. Extending network range
- Answer: B. Managing multiple access points**
-

45. What is the role of frequency channels in WLANs?

- A. Encrypting traffic
 - **B. Reducing interference between networks**
 - C. Assigning IP addresses
 - D. Improving range
- Answer: B. Reducing interference between networks**
-

46. What does WPA2 use for encryption?

- A. TKIP
 - **B. AES**
 - C. WEP
 - D. SSL
- Answer: B. AES**
-

47. Which device converts wireless signals into wired Ethernet?

- A. Wireless adapter
 - B. Repeater
 - **C. Wireless bridge**
 - D. Access point
- Answer: C. Wireless bridge**
-

48. What does RTS/CTS stand for in WLANs?

- **A. Request to Send/Clear to Send**
 - B. Ready to Signal/Complete to Signal
 - C. Repeater Transfer Signal/Completed Transfer Signal
 - D. Remote Transmission System/Clear Transmission System
- Answer: A. Request to Send/Clear to Send**
-

49. What is the maximum theoretical speed of 802.11ac?

- A. 433 Mbps
- B. 867 Mbps
- **C. 1.3 Gbps**
- D. 6.9 Gbps

Answer: C. 1.3 Gbps

50. What is an ESS in WLANs?

- A. Extended Security System
- **B. Extended Service Set**
- C. Enhanced Speed System
- D. Encrypted Service System

Answer: B. Extended Service Set

TOPIC 5 : WI-FI

1. What does Wi-Fi stand for?

- A. Wireless Fidelity
- **B. It does not officially stand for anything**
- C. Wireless Interface
- D. Wireless Frequency

Answer: B. It does not officially stand for anything

2. Which organization is responsible for certifying Wi-Fi devices?

- A. IEEE
- **B. Wi-Fi Alliance**
- C. IETF
- D. ICANN

Answer: B. Wi-Fi Alliance

3. Which IEEE standard is the basis for Wi-Fi technology?

- A. 802.3
- **B. 802.11**
- C. 802.15
- D. 802.16

Answer: B. 802.11

4. What frequency bands are commonly used in Wi-Fi networks?

- A. 900 MHz and 1.8 GHz
- **B. 2.4 GHz and 5 GHz**
- C. 4 GHz and 5.5 GHz
- D. 3 GHz and 6 GHz

Answer: B. 2.4 GHz and 5 GHz

5. What is the purpose of a Wi-Fi access point?

- A. Encrypting data packets
- **B. Providing wireless connectivity**
- C. Extending network range
- D. Assigning IP addresses

Answer: B. Providing wireless connectivity

6. What is the maximum theoretical speed of Wi-Fi 6 (802.11ax)?

- A. 4.8 Gbps
- **B. 9.6 Gbps**
- C. 1 Gbps
- D. 6 Gbps

Answer: B. 9.6 Gbps

7. What does WPA stand for in Wi-Fi security?

- A. Wireless Protection Algorithm
- **B. Wi-Fi Protected Access**
- C. Wireless Protected Authentication
- D. Wireless Protocol Access

Answer: B. Wi-Fi Protected Access

8. Which type of IP address is commonly assigned in Wi-Fi networks?

- A. Static IP
- **B. Dynamic IP**
- C. Private IP
- D. Public IP

Answer: B. Dynamic IP

9. What is an SSID in Wi-Fi networks?

- A. Security Service ID
- **B. Service Set Identifier**
- C. Signal Strength Identification
- D. Secure System Identifier

Answer: B. Service Set Identifier

10. What is the range of a typical indoor Wi-Fi network?

- A. 5-10 meters
- **B. 20-50 meters**
- C. 60-80 meters
- D. 100-150 meters

Answer: B. 20-50 meters

11. What does a Wi-Fi repeater do?

- A. Encrypts wireless traffic
- **B. Extends the coverage area of a Wi-Fi network**
- C. Converts Wi-Fi to wired Ethernet
- D. Reduces network latency

Answer: B. Extends the coverage area of a Wi-Fi network

12. What encryption standard does WPA2 use?

- A. TKIP
- **B. AES**
- C. DES
- D. RSA

Answer: B. AES

13. Which Wi-Fi standard introduced MIMO technology?

- A. 802.11g
- B. 802.11b
- **C. 802.11n**
- D. 802.11a

Answer: C. 802.11n

14. What is the purpose of Wi-Fi Protected Setup (WPS)?

- A. Increasing signal range
- **B. Simplifying the process of connecting devices**
- C. Enhancing encryption strength
- D. Assigning IP addresses

Answer: B. Simplifying the process of connecting devices

15. What is a hotspot in Wi-Fi?

- A. A secure wireless network
- **B. A location that provides Wi-Fi access**
- C. A repeater used for coverage
- D. A type of wireless protocol

Answer: B. A location that provides Wi-Fi access

16. What is the role of a Wi-Fi router?

- A. Encrypts all data packets
- **B. Connects devices to the internet via Wi-Fi**
- C. Assigns static IP addresses

- D. Extends network range

Answer: B. Connects devices to the internet via Wi-Fi

17. What does dual-band Wi-Fi mean?

- A. Uses two access points
- **B. Supports both 2.4 GHz and 5 GHz frequencies**
- C. Operates on two separate channels
- D. Encrypts data using two layers

Answer: B. Supports both 2.4 GHz and 5 GHz frequencies

18. What is the primary limitation of the 2.4 GHz band?

- A. Short range
- **B. Susceptibility to interference**
- C. Incompatibility with modern devices
- D. High energy consumption

Answer: B. Susceptibility to interference

19. What does the term "roaming" refer to in Wi-Fi networks?

- A. Encrypting wireless connections
- **B. Seamlessly moving between access points**
- C. Extending the signal range
- D. Assigning dynamic IP addresses

Answer: B. Seamlessly moving between access points

20. What is the function of DHCP in a Wi-Fi network?

- A. Encrypts data packets
- **B. Assigns IP addresses automatically**
- C. Increases signal strength
- D. Reduces latency

Answer: B. Assigns IP addresses automatically

21. What is the purpose of a MAC address in Wi-Fi networks?

- A. Encrypts data packets
- **B. Uniquely identifies a device**
- C. Measures signal strength
- D. Assigns IP addresses

Answer: B. Uniquely identifies a device

22. Which Wi-Fi standard is known as Wi-Fi 6?

- A. 802.11ac

- B. 802.11n
 - **C. 802.11ax**
 - D. 802.11g
- Answer: C. 802.11ax**
-

23. What is the maximum theoretical speed of 802.11ac (Wi-Fi 5)?

- A. 1.3 Gbps
 - **B. 6.9 Gbps**
 - C. 9.6 Gbps
 - D. 3.2 Gbps
- Answer: B. 6.9 Gbps**
-

24. What does OFDM stand for in Wi-Fi technology?

- A. Optimized Frequency Data Multiplexing
 - **B. Orthogonal Frequency Division Multiplexing**
 - C. Offset Frequency Data Management
 - D. Orthogonal Frequency Data Modulation
- Answer: B. Orthogonal Frequency Division Multiplexing**
-

25. What is the primary advantage of the 5 GHz Wi-Fi band over 2.4 GHz?

- A. Better range
 - **B. Higher speeds**
 - C. Greater device compatibility
 - D. Lower power consumption
- Answer: B. Higher speeds**
-

26. What is the role of beamforming in Wi-Fi networks?

- A. Splitting data streams
 - **B. Directing Wi-Fi signals toward devices**
 - C. Increasing encryption strength
 - D. Reducing network interference
- Answer: B. Directing Wi-Fi signals toward devices**
-

27. What does WPA3 offer compared to WPA2?

- A. Simpler setup
 - **B. Enhanced security**
 - C. Faster speeds
 - D. Greater range
- Answer: B. Enhanced security**
-

28. What is a mesh Wi-Fi network?

- A. A network with only one router
 - **B. A system of interconnected access points**
 - C. A temporary Wi-Fi hotspot
 - D. A high-speed mobile data network
- Answer: B. A system of interconnected access points**
-

29. What is the main function of a Wi-Fi extender?

- A. Encrypts network traffic
 - **B. Boosts Wi-Fi signal coverage**
 - C. Assigns IP addresses
 - D. Manages network congestion
- Answer: B. Boosts Wi-Fi signal coverage**
-

30. Which type of interference affects Wi-Fi signals most commonly?

- A. Light interference
 - B. Magnetic interference
 - **C. Radio frequency interference**
 - D. Physical obstruction interference
- Answer: C. Radio frequency interference**
-

31. What is the maximum channel width supported by Wi-Fi 6?

- A. 40 MHz
 - B. 80 MHz
 - **C. 160 MHz**
 - D. 320 MHz
- Answer: C. 160 MHz**
-

32. What does a captive portal do in a Wi-Fi network?

- **A. Requires users to authenticate via a web page**
 - B. Extends the range of the network
 - C. Encrypts network traffic
 - D. Automatically assigns IP addresses
- Answer: A. Requires users to authenticate via a web page**
-

33. What does RTS/CTS stand for in Wi-Fi?

- **A. Request to Send / Clear to Send**
 - B. Ready to Signal / Complete to Signal
 - C. Repeater Transmission Signal / Clear Transmission Signal
 - D. Remote Transfer System / Completed Transfer System
- Answer: A. Request to Send / Clear to Send**
-

34. What is a rogue access point?

- A. A backup access point
- B. A secure Wi-Fi router
- **C. An unauthorized access point**
- D. A device used for mesh networks

Answer: C. An unauthorized access point

35. What is the purpose of QoS in Wi-Fi networks?

- A. Improves encryption
 - **B. Prioritizes traffic for better performance**
 - C. Extends network range
 - D. Simplifies network setup
- Answer: B. Prioritizes traffic for better performance**
-

36. What does MU-MIMO stand for in Wi-Fi?

- **A. Multi-User Multiple Input Multiple Output**
 - B. Multiple Uplink Multiple Interference Management Output
 - C. Mobile User Multiple Input Multiplexing Output
 - D. Managed User Multipoint Integration Output
- Answer: A. Multi-User Multiple Input Multiple Output**
-

37. What is Wi-Fi Direct?

- A. A feature for extending Wi-Fi coverage
 - B. A secure way to connect to public hotspots
 - **C. A peer-to-peer wireless connection technology**
 - D. A protocol for device authentication
- Answer: C. A peer-to-peer wireless connection technology**
-

38. Which layer of the OSI model does Wi-Fi operate on?

- A. Physical layer only
 - **B. Both Physical and Data Link layers**
 - C. Network layer
 - D. Transport layer
- Answer: B. Both Physical and Data Link layers**
-

39. What is the primary function of a Wi-Fi antenna?

- A. Assigns dynamic IP addresses
 - **B. Sends and receives radio signals**
 - C. Boosts network speed
 - D. Encrypts network traffic
- Answer: B. Sends and receives radio signals**

40. What does the term "backhaul" mean in a Wi-Fi mesh network?

- A. Extending the network range
- B. Reducing latency
- **C. Communication between access points**
- D. Encrypting network traffic

Answer: C. Communication between access points

41. What is the maximum theoretical speed of Wi-Fi 4 (802.11n)?

- A. 100 Mbps
 - **B. 600 Mbps**
 - C. 1 Gbps
 - D. 300 Mbps
- Answer: B. 600 Mbps**
-

42. What is Wi-Fi calling?

- A. A VoIP service
 - **B. Making calls over a Wi-Fi network**
 - C. A feature of mesh networks
 - D. A secure calling protocol
- Answer: B. Making calls over a Wi-Fi network**
-

43. Which Wi-Fi frequency band has more non-overlapping channels?

- A. 2.4 GHz
 - **B. 5 GHz**
 - C. 6 GHz
 - D. 900 MHz
- Answer: B. 5 GHz**
-

44. What does "handover" mean in Wi-Fi?

- **A. Transitioning between access points without losing connection**
 - B. Encrypting traffic during device connection
 - C. Sharing bandwidth among devices
 - D. Assigning dynamic IP addresses
- Answer: A. Transitioning between access points without losing connection**
-

45. What is the role of Wi-Fi certification?

- A. Enhancing encryption protocols
- B. Extending signal range
- **C. Ensuring device interoperability**

- D. Assigning IP addresses
- Answer: C. Ensuring device interoperability**
-

46. What is a wireless SSID broadcast?

- A. A security protocol
 - **B. The transmission of the network name**
 - C. A method for encrypting traffic
 - D. A feature for mesh networking
- Answer: B. The transmission of the network name**
-

47. Which Wi-Fi protocol is best for IoT devices?

- A. 802.11ac
 - B. 802.11n
 - **C. 802.11ah (HaLow)**
 - D. 802.11ax
- Answer: C. 802.11ah (HaLow)**
-

48. What is the purpose of Wi-Fi 6E?

- A. Extending the range of Wi-Fi
 - B. Increasing compatibility with older devices
 - **C. Expanding Wi-Fi to the 6 GHz band**
 - D. Improving security features
- Answer: C. Expanding Wi-Fi to the 6 GHz band**
-

49. What is the maximum number of devices that can connect to a Wi-Fi router?

- A. 20
 - B. 50
 - **C. It varies depending on the router**
 - D. Unlimited
- Answer: C. It varies depending on the router**
-

50. What is the typical range of a Wi-Fi network outdoors?

- A. 10-50 meters
 - **B. 100-200 meters**
 - C. 300-500 meters
 - D. 1 km
- Answer: B. 100-200 meters**

TOPIC 6 : LAN TESTING

1. What does LAN stand for?

- A) Local Area Network
 - B) Large Area Network
 - C) Local Access Network
 - D) Long Area Network
- Answer:** A) Local Area Network

2. Which tool is commonly used for LAN cable testing?

- A) Oscilloscope
 - B) Cable Tester
 - C) Spectrum Analyzer
 - D) Multimeter
- Answer:** B) Cable Tester

3. What is the main purpose of LAN testing?

- A) To measure network speed
 - B) To verify cable integrity
 - C) To check router settings
 - D) To detect IP addresses
- Answer:** B) To verify cable integrity

4. Which of the following tests can a cable tester perform?

- A) Continuity Test
 - B) Signal Strength Test
 - C) Speed Test
 - D) Bandwidth Test
- Answer:** A) Continuity Test

5. What is the maximum cable length for a Cat 5e Ethernet cable?

- A) 50 meters
 - B) 100 meters
 - C) 200 meters
 - D) 300 meters
- Answer:** B) 100 meters

6. What does the term “link speed” refer to in a LAN environment?

- A) Data transfer rate
 - B) Voltage level
 - C) Frequency
 - D) Cable length
- Answer:** A) Data transfer rate

7. Which device in a LAN environment is responsible for forwarding data to the correct destination?

- A) Hub
 - B) Switch
 - C) Router
 - D) Modem
- Answer:** B) Switch

8. Which of the following is used to test network performance and latency in a LAN?

- A) Ping Test
- B) Cable Tester
- C) Multimeter

D) Protocol Analyzer
Answer: A) Ping Test

9. What is the purpose of the "loopback" test in LAN testing?

A) To test network speed
B) To check for signal loss
C) To test the integrity of the network interface card (NIC)
D) To measure the bandwidth
Answer: C) To test the integrity of the network interface card (NIC)

10. Which type of LAN cable is most commonly used in modern networks?

A) Coaxial Cable
B) Fiber Optic Cable
C) Unshielded Twisted Pair (UTP) Cable
D) Shielded Twisted Pair (STP) Cable
Answer: C) Unshielded Twisted Pair (UTP) Cable

11. What does the acronym "IP" stand for in networking?

A) Internet Protocol
B) Internet Point
C) Internal Protocol
D) Integrated Packet
Answer: A) Internet Protocol

12. In LAN testing, what does the term "attenuation" refer to?

A) Signal loss
B) Signal strength
C) Signal interference
D) Signal clarity
Answer: A) Signal loss

13. What is the typical voltage used in Ethernet cables for signaling?

A) 3V
B) 5V
C) 12V
D) 48V
Answer: D) 48V

14. Which device would you use to check the signal quality of a LAN connection?

A) Cable Tester
B) Network Analyzer
C) Oscilloscope
D) Multimeter
Answer: B) Network Analyzer

15. What does a "green" LED light on a network interface card (NIC) typically indicate?

A) Cable failure
B) Link activity
C) Signal interference
D) Power failure
Answer: B) Link activity

16. What is a common reason for slow network speeds in a LAN?

- A) Outdated hardware
 - B) High network traffic
 - C) Signal interference
 - D) All of the above
- Answer:** D) All of the above

17. Which of the following is used to test for a “split pair” in Ethernet cables?

- A) Tone Generator
 - B) Cable Tester
 - C) Network Analyzer
 - D) Multimeter
- Answer:** B) Cable Tester

18. What kind of testing does a Time Domain Reflectometer (TDR) perform?

- A) Cable Length Measurement
 - B) Signal Reflection Measurement
 - C) Bandwidth Testing
 - D) Network Topology Mapping
- Answer:** B) Signal Reflection Measurement

19. Which layer of the OSI model is responsible for data link testing in a LAN?

- A) Physical Layer
 - B) Data Link Layer
 - C) Network Layer
 - D) Transport Layer
- Answer:** B) Data Link Layer

20. What is the maximum transmission distance of a single-mode fiber optic cable?

- A) 10 km
 - B) 50 km
 - C) 100 km
 - D) 500 km
- Answer:** B) 50 km

21. What is the function of a crossover cable in a LAN?

- A) To connect devices of the same type
 - B) To extend the network range
 - C) To connect a PC to a router
 - D) To connect different types of networks
- Answer:** A) To connect devices of the same type

22. What does "PoE" stand for in LAN testing?

- A) Power over Ethernet
 - B) Power of Equipment
 - C) Point of Entry
 - D) Protocol over Ethernet
- Answer:** A) Power over Ethernet

23. What is the primary purpose of a patch panel in LANs?

- A) To connect devices within a LAN
- B) To distribute network traffic evenly
- C) To provide wireless connectivity

D) To manage and organize network cables
Answer: D) To manage and organize network cables

24. Which of the following network topologies is the easiest to troubleshoot?

- A) Star Topology
- B) Bus Topology
- C) Ring Topology
- D) Mesh Topology

Answer: A) Star Topology

25. Which tool is used to detect network congestion in a LAN?

- A) Network Analyzer
- B) Cable Tester
- C) Oscilloscope
- D) Multimeter

Answer: A) Network Analyzer

26. What is the role of an Ethernet switch in a LAN?

- A) Routing data to different networks
- B) Repeating signals to extend distance
- C) Directing data traffic within a local network
- D) Providing internet access

Answer: C) Directing data traffic within a local network

27. What can cause network interference during LAN testing?

- A) Electrical equipment
- B) Poor cable shielding
- C) Distance between devices
- D) All of the above

Answer: D) All of the above

28. What type of testing would be done using a "bit error rate" in a LAN?

- A) Data integrity testing
- B) Voltage testing
- C) Speed testing
- D) Cable continuity testing

Answer: A) Data integrity testing

29. What does "BNC" stand for in networking?

- A) Basic Network Cable
- B) British Network Connector
- C) Bayonet Neill–Concelman connector
- D) Binary Network Cable

Answer: C) Bayonet Neill–Concelman connector

30. Which of the following is the best practice when testing LAN cables for connectivity?

- A) Test only the physical connection
- B) Test for both wire pairing and signal integrity
- C) Test for signal interference only
- D) Test for signal strength only

Answer: B) Test for both wire pairing and signal integrity

31. What type of cable is used for long-distance network connections in LANs?

- A) Coaxial Cable
 - B) Fiber Optic Cable
 - C) UTP Cable
 - D) STP Cable
- Answer:** B) Fiber Optic Cable

32. What does the "ping" command in a network test verify?

- A) Cable continuity
 - B) Network latency
 - C) Signal strength
 - D) Router configuration
- Answer:** B) Network latency

33. How can network congestion be minimized in a LAN?

- A) Adding more network cables
 - B) Using faster routers
 - C) Implementing network segmentation
 - D) Using fiber optic cables
- Answer:** C) Implementing network segmentation

34. Which of the following tools would you use to check the health of network cables?

- A) Cable Tester
 - B) Spectrum Analyzer
 - C) Oscilloscope
 - D) Protocol Analyzer
- Answer:** A) Cable Tester

35. In a fiber optic connection, what is used to test the power levels?

- A) Light Meter
 - B) Cable Tester
 - C) Signal Generator
 - D) Network Analyzer
- Answer:** A) Light Meter

36. Which of the following tests is used to determine the speed of a LAN?

- A) Ping Test
 - B) Speed Test
 - C) Cable Continuity Test
 - D) Signal Interference Test
- Answer:** B) Speed Test

37. Which of the following devices helps in analyzing data packets in a LAN?

- A) Hub
 - B) Switch
 - C) Protocol Analyzer
 - D) Router
- Answer:** C) Protocol Analyzer

38. What is the purpose of an Ethernet crossover cable in LAN testing?

- A) To increase network speed
- B) To connect different network segments
- C) To connect two computers directly

D) To extend the range of the network
Answer: C) To connect two computers directly

39. What is the primary purpose of using a TDR (Time Domain Reflectometer) in LAN testing?

- A) Measuring cable length
- B) Detecting cable faults
- C) Checking for network congestion
- D) Measuring signal speed

Answer: B) Detecting cable faults

40. In which of the following cases would you use a loopback adapter in LAN testing?

- A) To test the router configuration
- B) To check network speed
- C) To test a network card
- D) To measure packet loss

Answer: C) To test a network card

41. Which cable type is used for shielding against electromagnetic interference (EMI) in a LAN?

- A) STP (Shielded Twisted Pair)
- B) UTP (Unshielded Twisted Pair)
- C) Fiber Optic Cable
- D) Coaxial Cable

Answer: A) STP (Shielded Twisted Pair)

42. Which of the following protocols is used to test connectivity between devices in a LAN?

- A) HTTP
- B) FTP
- C) ICMP
- D) TCP

Answer: C) ICMP

43. What is a VLAN in a network?

- A) Virtual Local Area Network
- B) Virtual Link Active Node
- C) Variable Local Area Network
- D) Visual Local Area Node

Answer: A) Virtual Local Area Network

44. What does "DHCP" stand for in networking?

- A) Dynamic Host Configuration Protocol
- B) Dynamic Home Configuration Protocol
- C) Distributed Host Configuration Protocol
- D) Direct Host Configuration Protocol

Answer: A) Dynamic Host Configuration Protocol

45. Which of the following devices is typically used for diagnosing network issues in a LAN?

- A) Cable Tester
- B) Protocol Analyzer
- C) Router
- D) Hub

Answer: B) Protocol Analyzer

46. What is the function of a network switch in LAN testing?

- A) It routes data between different networks
 - B) It repeats data signals over long distances
 - C) It connects devices within a LAN and manages data traffic
 - D) It manages wireless connectivity
- Answer:** C) It connects devices within a LAN and manages data traffic

47. Which type of fiber optic cable is used for short-range connections in LANs?

- A) Single-mode fiber
 - B) Multi-mode fiber
 - C) STP fiber
 - D) UTP fiber
- Answer:** B) Multi-mode fiber

48. What is the purpose of the "ARP" protocol in a LAN?

- A) To route data between networks
 - B) To resolve IP addresses to MAC addresses
 - C) To control data flow
 - D) To check network security
- Answer:** B) To resolve IP addresses to MAC addresses

49. What does a "collisions" test in LAN testing measure?

- A) Signal loss
 - B) Data packet conflicts on the network
 - C) Voltage fluctuation
 - D) Network speed
- Answer:** B) Data packet conflicts on the network

50. In which situation would a "fiber optic" cable be most suitable for LAN testing?

- A) For short-distance, high-speed connections
 - B) For high resistance to electrical interference
 - C) For long-distance, high-bandwidth requirements
 - D) All of the above
- Answer:** D) All of the above

TOPIC 7 : LAN Proxy Server

1. What is the primary function of a proxy server in a LAN?

- A) To forward requests between clients and servers
 - B) To store data for clients
 - C) To secure the LAN from external threats
 - D) To manage IP addresses
- Answer:** A) To forward requests between clients and servers

2. Which of the following is a common use of a proxy server in a LAN?

- A) Improving network security
 - B) Distributing bandwidth
 - C) Assigning IP addresses
 - D) Managing user authentication
- Answer:** A) Improving network security

3. What type of proxy server works by forwarding requests from clients to the internet?

- A) Transparent Proxy
- B) Reverse Proxy

- C) Forward Proxy
 - D) Open Proxy
- Answer:** C) Forward Proxy

4. Which of the following is NOT a feature of a proxy server?

- A) Caching content
 - B) Encrypting communication
 - C) Filtering web traffic
 - D) Blocking external devices
- Answer:** D) Blocking external devices

5. A proxy server can help improve the performance of a LAN by performing which function?

- A) Reducing bandwidth
 - B) Caching frequently accessed content
 - C) Encrypting communication
 - D) Blocking external connections
- Answer:** B) Caching frequently accessed content

6. Which of the following types of proxy servers is used to provide access to a LAN from the internet?

- A) Reverse Proxy
 - B) Forward Proxy
 - C) Transparent Proxy
 - D) Open Proxy
- Answer:** A) Reverse Proxy

7. Which protocol does a proxy server typically use to communicate with clients?

- A) HTTP
 - B) FTP
 - C) TCP/IP
 - D) SNMP
- Answer:** A) HTTP

8. What does a proxy server typically cache?

- A) User credentials
 - B) Web pages and files
 - C) Network IP addresses
 - D) Email messages
- Answer:** B) Web pages and files

9. Which of the following benefits does a proxy server provide in terms of security?

- A) Hiding the client's IP address
 - B) Encrypting emails
 - C) Routing data through secure networks
 - D) Blocking malware on the network
- Answer:** A) Hiding the client's IP address

10. In a proxy server setup, which device typically handles the client's request for a web page?

- A) Client machine
 - B) Proxy server
 - C) Web server
 - D) DNS server
- Answer:** B) Proxy server

11. A proxy server configured to allow only specific websites is an example of what kind of function?

- A) Web caching
- B) Web filtering
- C) Load balancing
- D) Bandwidth management

Answer: B) Web filtering

12. Which of the following is a disadvantage of using a proxy server?

- A) Increased bandwidth usage
- B) Centralized point of failure
- C) Improved security
- D) Faster access to resources

Answer: B) Centralized point of failure

13. Which proxy server type would you use to provide load balancing for multiple web servers?

- A) Forward Proxy
- B) Reverse Proxy
- C) Transparent Proxy
- D) Open Proxy

Answer: B) Reverse Proxy

14. Which of the following is true about transparent proxy servers?

- A) They require special client-side configuration
- B) They do not modify requests or responses
- C) They need to be explicitly configured by the user
- D) They are used for encryption purposes

Answer: B) They do not modify requests or responses

15. A reverse proxy server sits between the internet and which of the following?

- A) DNS server
- B) Web server
- C) Client machines
- D) Switches

Answer: B) Web server

16. In which scenario would a proxy server be used for content filtering?

- A) Blocking access to specific websites
- B) Improving data transmission speed
- C) Encrypting user data
- D) Ensuring network redundancy

Answer: A) Blocking access to specific websites

17. What is one of the primary security roles of a proxy server?

- A) Hiding user credentials
- B) Preventing malware from entering the LAN
- C) Providing encryption for email communications
- D) Blocking access to unauthorized ports

Answer: B) Preventing malware from entering the LAN

18. Which of the following is an advantage of using a proxy server for web traffic?

- A) Increased server load
- B) Decreased network security

- C) Reduced latency
 - D) Better access control
- Answer:** D) Better access control

19. A proxy server that only works with HTTP requests is known as what?

- A) FTP Proxy
 - B) HTTP Proxy
 - C) SOCKS Proxy
 - D) DNS Proxy
- Answer:** B) HTTP Proxy

20. A forward proxy is typically used to do which of the following?

- A) Hide the identity of clients accessing the internet
 - B) Load balance web traffic between servers
 - C) Provide access to an intranet from outside
 - D) Cache web content for all internal clients
- Answer:** A) Hide the identity of clients accessing the internet

21. What is one advantage of using a proxy server for web traffic caching?

- A) It helps with web page encryption
 - B) It reduces bandwidth consumption
 - C) It creates additional security vulnerabilities
 - D) It ensures more accurate DNS resolution
- Answer:** B) It reduces bandwidth consumption

22. What does a proxy server do with web requests?

- A) It denies all requests for external content
 - B) It forwards requests from clients to servers
 - C) It stores all data for later access
 - D) It filters out unauthorized traffic from the network
- Answer:** B) It forwards requests from clients to servers

23. Which type of proxy server would you use if you wanted to provide an additional layer of security between your LAN and the internet?

- A) Forward Proxy
 - B) Transparent Proxy
 - C) Reverse Proxy
 - D) Open Proxy
- Answer:** C) Reverse Proxy

24. Which of the following best describes the caching function of a proxy server?

- A) Storing data on a client's machine
 - B) Storing frequently accessed data to speed up access
 - C) Encrypting all user data
 - D) Blocking unauthorized websites
- Answer:** B) Storing frequently accessed data to speed up access

25. What is a disadvantage of proxy server caching?

- A) Increased network speed
 - B) Outdated or incorrect data can be delivered to clients
 - C) Improved data security
 - D) No impact on network performance
- Answer:** B) Outdated or incorrect data can be delivered to clients

26. How does a proxy server enhance privacy on a LAN?

- A) By masking the client's real IP address
- B) By blocking all external internet access
- C) By hiding sensitive files on the server
- D) By using encryption protocols

Answer: A) By masking the client's real IP address

27. Which of the following is an example of a proxy server feature aimed at improving bandwidth management?

- A) Blocking access to specific websites
- B) Caching commonly accessed web pages
- C) Encrypting user data
- D) Authenticating users

Answer: B) Caching commonly accessed web pages

28. What type of proxy server is typically used to provide a single point of contact for multiple web servers in a LAN?

- A) Transparent Proxy
- B) Forward Proxy
- C) Reverse Proxy
- D) Open Proxy

Answer: C) Reverse Proxy

29. What is a key function of a proxy server in a corporate environment?

- A) Managing web traffic routing
- B) Encrypting all incoming web data
- C) Blocking email access
- D) Providing access to external VPNs

Answer: A) Managing web traffic routing

30. What type of proxy server is typically used in online anonymity services?

- A) Transparent Proxy
- B) Anonymous Proxy
- C) Reverse Proxy
- D) SOCKS Proxy

Answer: B) Anonymous Proxy

31. Which of the following best describes a SOCKS proxy?

- A) A proxy that works only with HTTP traffic
- B) A proxy used to anonymize internet browsing
- C) A proxy server that caches web content
- D) A proxy for forwarding email traffic

Answer: B) A proxy used to anonymize internet browsing

32. What does a proxy server need to modify in order to forward web traffic properly?

- A) The DNS settings
- B) The request headers
- C) The IP addresses
- D) The network configuration

Answer: B) The request headers

33. What is one use case for a proxy server in a corporate LAN environment?

- A) Providing direct access to the internet for all devices
- B) Filtering and blocking unwanted websites
- C) Encrypting all web traffic
- D) Storing personal user data

Answer: B) Filtering and blocking unwanted websites

34. In a proxy server setup, which of the following components can be secured by encrypting traffic?

- A) Client-to-server connections
- B) Proxy-to-web server connections
- C) Only the DNS queries
- D) All of the above

Answer: B) Proxy-to-web server connections

35. What type of proxy server is designed to be invisible to users?

- A) Transparent Proxy
- B) Reverse Proxy
- C) Forward Proxy
- D) Open Proxy

Answer: A) Transparent Proxy

36. What is the main advantage of using a proxy server for security in a network?

- A) Directing all traffic through a secure tunnel
- B) Blocking all incoming traffic
- C) Providing encryption for external traffic
- D) Masking the identity of internal users

Answer: D) Masking the identity of internal users

37. How does a proxy server provide anonymity to its users?

- A) By hiding the client's IP address
- B) By blocking all incoming traffic
- C) By encrypting all data
- D) By allowing direct access to external sites

Answer: A) By hiding the client's IP address

38. What kind of proxy server is commonly used in corporate environments to distribute web traffic efficiently across multiple servers?

- A) Forward Proxy
- B) Reverse Proxy
- C) Transparent Proxy
- D) SOCKS Proxy

Answer: B) Reverse Proxy

39. Which of the following is an important security function of a proxy server in a LAN?

- A) Increasing web traffic speed
- B) Protecting against denial-of-service attacks
- C) Managing DNS resolution
- D) Encrypting email communication

Answer: B) Protecting against denial-of-service attacks

40. In which scenario would a proxy server be used for filtering content based on the type of data?

- A) Blocking streaming services
- B) Blocking specific file types from being downloaded
- C) Providing access to private files

D) Encrypting internet traffic

Answer: B) Blocking specific file types from being downloaded

41. How does a proxy server support a distributed network?

A) By providing load balancing and caching for web servers

B) By blocking all external communication

C) By encrypting user passwords

D) By sharing the same IP address for all devices

Answer: A) By providing load balancing and caching for web servers

42. What happens if a proxy server becomes unavailable in a network setup?

A) All web traffic will be redirected to another proxy

B) Clients may lose internet connectivity or experience slow speeds

C) The network will automatically switch to a new DNS server

D) All proxy functions will be disabled

Answer: B) Clients may lose internet connectivity or experience slow speeds

43. What is the role of a proxy server in preventing a Distributed Denial of Service (DDoS) attack?

A) By blocking all incoming requests

B) By hiding the internal network structure and filtering malicious requests

C) By encrypting the web traffic

D) By storing data temporarily to reduce server load

Answer: B) By hiding the internal network structure and filtering malicious requests

44. Which of the following protocols does a proxy server use to communicate with clients?

A) HTTPS

B) HTTP

C) FTP

D) SNMP

Answer: B) HTTP

45. What type of proxy is used to forward requests from a client inside a corporate LAN to the internet?

A) Reverse Proxy

B) Forward Proxy

C) Transparent Proxy

D) Web Proxy

Answer: B) Forward Proxy

46. What does a proxy server do when a client requests a cached web page?

A) It forwards the request to the original web server

B) It directly serves the cached page from the cache

C) It redirects the request to a backup server

D) It encrypts the page content

Answer: B) It directly serves the cached page from the cache

47. What is the main difference between a forward proxy and a reverse proxy?

A) A forward proxy is used to provide access to the internet, while a reverse proxy is used to protect web servers.

B) A forward proxy is used for content filtering, while a reverse proxy is for caching.

C) A forward proxy hides a user's IP address, while a reverse proxy hides the web server's IP address.

D) There is no difference between them.

Answer: A) A forward proxy is used to provide access to the internet, while a reverse proxy is used to protect web servers.

48. What kind of proxy server can be used to provide secure communication between clients and a web server over the internet?

- A) Forward Proxy
- B) Reverse Proxy
- C) Transparent Proxy
- D) Open Proxy

Answer: B) Reverse Proxy

49. What is the primary role of a transparent proxy?

- A) To increase network bandwidth
- B) To block malicious traffic
- C) To cache web content without altering the client's request
- D) To provide encrypted connections

Answer: C) To cache web content without altering the client's request

50. In a corporate environment, why would you use a proxy server for content filtering?

- A) To allow all internet traffic unrestricted access
- B) To prevent access to inappropriate or non-business-related websites
- C) To increase the speed of internet browsing
- D) To create a secure connection for mobile devices

Answer: B) To prevent access to inappropriate or non-business-related websites

TOPIC 8 : OSI Layers

1. What does OSI stand for in networking?

- A) Open Systems Interconnection
- B) Open Source Interface
- C) Operating System Interface
- D) Open Source Interconnection

Answer: A) Open Systems Interconnection

2. How many layers are there in the OSI model?

- A) 5
- B) 7
- C) 9
- D) 10

Answer: B) 7

3. Which layer of the OSI model is responsible for error detection and correction?

- A) Application Layer
- B) Presentation Layer
- C) Transport Layer
- D) Data Link Layer

Answer: D) Data Link Layer

4. Which of the following is the main function of the Physical Layer in the OSI model?

- A) Ensuring reliable data transmission
- B) Providing network access
- C) Transmitting raw bit streams over a physical medium
- D) Compressing data

Answer: C) Transmitting raw bit streams over a physical medium

5. Which layer of the OSI model is responsible for end-to-end communication?

- A) Transport Layer
 - B) Network Layer
 - C) Session Layer
 - D) Application Layer
- Answer:** A) Transport Layer

6. Which OSI layer is responsible for routing data packets?

- A) Network Layer
 - B) Transport Layer
 - C) Data Link Layer
 - D) Physical Layer
- Answer:** A) Network Layer

7. Which layer of the OSI model provides encryption and decryption of data?

- A) Application Layer
 - B) Presentation Layer
 - C) Session Layer
 - D) Transport Layer
- Answer:** B) Presentation Layer

8. At which layer of the OSI model do routers operate?

- A) Network Layer
 - B) Data Link Layer
 - C) Transport Layer
 - D) Physical Layer
- Answer:** A) Network Layer

9. Which of the following is the main function of the Session Layer in the OSI model?

- A) Managing data flow control
 - B) Establishing, maintaining, and terminating communication sessions
 - C) Routing data packets
 - D) Encrypting data
- Answer:** B) Establishing, maintaining, and terminating communication sessions

10. Which OSI layer provides network services to the application layer?

- A) Transport Layer
 - B) Network Layer
 - C) Session Layer
 - D) Presentation Layer
- Answer:** A) Transport Layer

11. Which layer is responsible for ensuring reliable communication between two devices over a network?

- A) Application Layer
 - B) Network Layer
 - C) Transport Layer
 - D) Data Link Layer
- Answer:** C) Transport Layer

12. Which layer in the OSI model is responsible for data compression and decompression?

- A) Application Layer
- B) Presentation Layer
- C) Transport Layer

D) Network Layer

Answer: B) Presentation Layer

13. Which OSI layer provides the interface between the application and the network?

A) Network Layer

B) Application Layer

C) Transport Layer

D) Data Link Layer

Answer: B) Application Layer

14. Which of the following is not a function of the Data Link Layer?

A) Error detection and correction

B) Framing

C) Addressing (MAC Address)

D) Routing

Answer: D) Routing

15. Which OSI layer ensures that data is delivered without errors, in the correct sequence, and without duplication?

A) Network Layer

B) Transport Layer

C) Data Link Layer

D) Application Layer

Answer: B) Transport Layer

16. Which layer of the OSI model is responsible for segmentation and reassembly of data?

A) Data Link Layer

B) Transport Layer

C) Application Layer

D) Network Layer

Answer: B) Transport Layer

17. At which OSI layer do switches operate?

A) Data Link Layer

B) Network Layer

C) Transport Layer

D) Application Layer

Answer: A) Data Link Layer

18. The OSI model defines how many layers for communication?

A) 5

B) 6

C) 7

D) 8

Answer: C) 7

19. Which of the following devices operates at the OSI model's Physical Layer?

A) Router

B) Switch

C) Hub

D) Gateway

Answer: C) Hub

20. In the OSI model, what is the role of the Network Layer?

- A) Provides a user interface to access the network
- B) Manages data flow control
- C) Routes packets across the network
- D) Transmits raw data bits

Answer: C) Routes packets across the network

21. Which OSI layer is responsible for providing logical addressing?

- A) Transport Layer
- B) Network Layer
- C) Data Link Layer
- D) Physical Layer

Answer: B) Network Layer

22. What is the primary function of the Application Layer in the OSI model?

- A) Encrypt data
- B) Interface between the application and the network
- C) Route data packets
- D) Provide end-to-end error correction

Answer: B) Interface between the application and the network

23. Which OSI layer is responsible for creating a reliable end-to-end communication path?

- A) Transport Layer
- B) Network Layer
- C) Application Layer
- D) Session Layer

Answer: A) Transport Layer

24. Which of the following is NOT a responsibility of the Presentation Layer in the OSI model?

- A) Data translation
- B) Compression
- C) Encryption
- D) Routing

Answer: D) Routing

25. What is the main responsibility of the Transport Layer in the OSI model?

- A) To encrypt data for secure communication
- B) To handle the routing of packets
- C) To ensure complete and accurate delivery of data
- D) To establish communication sessions

Answer: C) To ensure complete and accurate delivery of data

26. Which OSI layer is involved in converting digital signals to analog signals for transmission over physical media?

- A) Network Layer
- B) Transport Layer
- C) Physical Layer
- D) Data Link Layer

Answer: C) Physical Layer

27. What is the purpose of the Session Layer in the OSI model?

- A) It manages the establishment, maintenance, and termination of communication between devices.
 - B) It defines how data should be transmitted physically.
 - C) It ensures reliable data transfer.
 - D) It handles end-to-end error recovery.
- Answer:** A) It manages the establishment, maintenance, and termination of communication between devices.

28. Which of the following is an example of a device that operates at the Data Link Layer?

- A) Hub
 - B) Switch
 - C) Router
 - D) Bridge
- Answer:** B) Switch

29. Which OSI layer is responsible for the logical addressing of packets?

- A) Network Layer
 - B) Data Link Layer
 - C) Application Layer
 - D) Transport Layer
- Answer:** A) Network Layer

30. The OSI model's Transport Layer is responsible for which of the following?

- A) Error checking
 - B) Routing
 - C) Data encryption
 - D) Establishing communication sessions
- Answer:** A) Error checking

31. Which OSI layer determines how data is transmitted over the network?

- A) Network Layer
 - B) Data Link Layer
 - C) Physical Layer
 - D) Transport Layer
- Answer:** C) Physical Layer

32. At which layer of the OSI model are IP addresses used?

- A) Network Layer
 - B) Data Link Layer
 - C) Transport Layer
 - D) Application Layer
- Answer:** A) Network Layer

33. Which of the following functions does the Network Layer perform?

- A) Error detection
 - B) Data segmentation
 - C) Routing and forwarding of packets
 - D) Data encryption
- Answer:** C) Routing and forwarding of packets

34. Which of the following is a function of the Application Layer?

- A) Encryption and compression
- B) Providing error control
- C) Providing user interface to network services

D) Routing packets

Answer: C) Providing user interface to network services

35. What type of data does the Transport Layer handle?

A) Raw data

B) Segmented data

C) Routed data

D) Encrypted data

Answer: B) Segmented data

36. In the OSI model, which layer defines the physical characteristics of the network medium?

A) Application Layer

B) Physical Layer

C) Data Link Layer

D) Transport Layer

Answer: B) Physical Layer

37. What does the Data Link Layer use to address devices on the network?

A) IP addresses

B) MAC addresses

C) Port numbers

D) URLs

Answer: B) MAC addresses

38. What function does the Transport Layer perform in the OSI model?

A) Ensuring reliable delivery of data

B) Encrypting data

C) Translating data formats

D) Routing data packets

Answer: A) Ensuring reliable delivery of data

39. What is the main responsibility of the Presentation Layer?

A) Data encryption and compression

B) Establishing sessions

C) Routing packets

D) Data segmentation

Answer: A) Data encryption and compression

40. Which of the following is a common example of a device that operates at the OSI's Network Layer?

A) Switch

B) Hub

C) Router

D) Bridge

Answer: C) Router

41. The Session Layer manages which aspect of communication?

A) Encryption

B) Addressing

C) Connection establishment, maintenance, and termination

D) Data flow control

Answer: C) Connection establishment, maintenance, and termination

42. What is the role of the Physical Layer in the OSI model?

- A) Establishing, maintaining, and terminating communication sessions
 - B) Converting data into signals for transmission over the network medium
 - C) Routing and forwarding data packets
 - D) Providing error-free delivery of data
- Answer:** B) Converting data into signals for transmission over the network medium

43. Which OSI layer is responsible for packet switching?

- A) Network Layer
 - B) Data Link Layer
 - C) Transport Layer
 - D) Application Layer
- Answer:** A) Network Layer

44. Which layer handles the representation and translation of data?

- A) Network Layer
 - B) Presentation Layer
 - C) Data Link Layer
 - D) Transport Layer
- Answer:** B) Presentation Layer

45. Which OSI layer is responsible for addressing and delivering data to specific devices?

- A) Data Link Layer
 - B) Physical Layer
 - C) Application Layer
 - D) Network Layer
- Answer:** D) Network Layer

46. What does the Application Layer use to access the network?

- A) Ports
 - B) IP addresses
 - C) DNS
 - D) MAC addresses
- Answer:** A) Ports

47. Which OSI layer provides flow control?

- A) Data Link Layer
 - B) Transport Layer
 - C) Network Layer
 - D) Physical Layer
- Answer:** B) Transport Layer

48. At which OSI layer does data encryption take place?

- A) Session Layer
 - B) Presentation Layer
 - C) Network Layer
 - D) Transport Layer
- Answer:** B) Presentation Layer

49. Which OSI layer is primarily responsible for routing data across the network?

- A) Transport Layer
- B) Network Layer
- C) Data Link Layer

D) Physical Layer

Answer: B) Network Layer

50. Which OSI layer is responsible for creating and terminating sessions between applications?

A) Application Layer

B) Transport Layer

C) Session Layer

D) Network Layer

Answer: C) Session Layer

TOPIC 9 : Network protection and security

1. What is the primary purpose of a firewall in network security?

A) To prevent unauthorized access to or from a private network

B) To provide secure DNS resolution

C) To encrypt all data transmission

D) To monitor network traffic

Answer: A) To prevent unauthorized access to or from a private network

2. Which of the following is a common type of firewall?

A) Application firewall

B) Physical firewall

C) Cloud firewall

D) Software firewall

Answer: A) Application firewall

3. What does VPN stand for in networking?

A) Virtual Protocol Network

B) Virtual Private Network

C) Verified Private Network

D) Virtual Public Network

Answer: B) Virtual Private Network

4. What is the primary function of a VPN?

A) To encrypt data and provide secure remote access

B) To monitor network traffic for intrusions

C) To prevent denial-of-service attacks

D) To block suspicious websites

Answer: A) To encrypt data and provide secure remote access

5. Which of the following protocols is commonly used to secure web traffic?

A) HTTP

B) HTTPS

C) FTP

D) SNMP

Answer: B) HTTPS

6. What is an intrusion detection system (IDS)?

A) A tool that blocks unauthorized network access

B) A system that identifies malicious network activity

C) A firewall that protects the network

D) A program that encrypts communication

Answer: B) A system that identifies malicious network activity

7. Which of the following is an example of a denial-of-service (DoS) attack?

- A) Phishing
- B) Spoofing
- C) Flooding a network with excessive traffic
- D) Data theft

Answer: C) Flooding a network with excessive traffic

8. What does the term "phishing" refer to in network security?

- A) The act of flooding a network with traffic
- B) The act of trying to steal sensitive information by posing as a trustworthy entity
- C) The unauthorized access to a network
- D) The illegal use of network resources

Answer: B) The act of trying to steal sensitive information by posing as a trustworthy entity

9. What is the role of encryption in network security?

- A) To ensure data is transferred without delay
- B) To protect data from unauthorized access
- C) To speed up data transmission
- D) To monitor network activity

Answer: B) To protect data from unauthorized access

10. What is the primary function of a router in network security?

- A) To filter and block malicious traffic
- B) To route data between different networks
- C) To monitor the network for intrusions
- D) To encrypt data during transmission

Answer: B) To route data between different networks

11. What type of attack involves a hacker intercepting and altering communication between two parties?

- A) Man-in-the-middle attack
- B) Denial-of-service attack
- C) Phishing attack
- D) Spoofing attack

Answer: A) Man-in-the-middle attack

12. What is a VPN tunnel?

- A) A secure connection between two networks or devices over the internet
- B) A device that encrypts network traffic
- C) A server that stores VPN configuration data
- D) A method of routing traffic over a public network

Answer: A) A secure connection between two networks or devices over the internet

13. Which of the following is a method used to protect passwords in a network?

- A) Encryption
- B) Compression
- C) Fragmentation
- D) Tunneling

Answer: A) Encryption

14. What is a key characteristic of a public key infrastructure (PKI)?

- A) It uses symmetric encryption for securing data
- B) It manages digital certificates and encryption keys

- C) It allows unlimited internet access
 - D) It disables data compression
- Answer:** B) It manages digital certificates and encryption keys

15. What does an antivirus program primarily protect against?

- A) Malware and viruses
- B) Unauthorized network access
- C) Data loss during transmission
- D) Encryption breaches

Answer: A) Malware and viruses

16. What does multi-factor authentication (MFA) require?

- A) The use of one password for security
- B) Multiple methods of verifying identity
- C) Encryption of all communications
- D) Disabling all accounts after a set period of time

Answer: B) Multiple methods of verifying identity

17. Which security measure prevents unauthorized access by verifying the identity of users or devices?

- A) Encryption
- B) Authentication
- C) Authorization
- D) Auditing

Answer: B) Authentication

18. What is a DDoS attack?

- A) A method of securing data during transfer
- B) An attack that uses multiple systems to flood a target with traffic
- C) A method of securing user identities
- D) An attack designed to alter user credentials

Answer: B) An attack that uses multiple systems to flood a target with traffic

19. Which type of security is responsible for ensuring that only authorized users can access data?

- A) Integrity
- B) Availability
- C) Confidentiality
- D) Authentication

Answer: C) Confidentiality

20. What is a vulnerability scanner used for in network security?

- A) To monitor network traffic in real-time
- B) To detect and identify potential security weaknesses
- C) To block unauthorized access
- D) To establish secure remote connections

Answer: B) To detect and identify potential security weaknesses

21. Which of the following describes a "zero-day" exploit?

- A) An attack targeting a known vulnerability that has been patched
- B) An attack that exploits a vulnerability before the vendor releases a fix
- C) A security measure to prevent attacks
- D) An attack that involves overwhelming a server with traffic

Answer: B) An attack that exploits a vulnerability before the vendor releases a fix

22. What is a security policy?

- A) A program that automatically detects network intrusions
 - B) A set of guidelines and procedures for maintaining network security
 - C) A device that blocks unauthorized access
 - D) A password manager used for data protection
- Answer:** B) A set of guidelines and procedures for maintaining network security

23. Which of the following is a common type of malicious software?

- A) Firewall
 - B) Virus
 - C) Router
 - D) Switch
- Answer:** B) Virus

24. What is a primary reason for implementing network segmentation in security?

- A) To improve network performance
 - B) To restrict access to sensitive data and limit the impact of security breaches
 - C) To increase the number of connected devices
 - D) To improve encryption
- Answer:** B) To restrict access to sensitive data and limit the impact of security breaches

25. Which of the following protocols is commonly used to secure email communication?

- A) SMTP
 - B) IMAP
 - C) POP3
 - D) S/MIME
- Answer:** D) S/MIME

26. What is the role of a proxy server in network security?

- A) To block malicious traffic
 - B) To route data between networks
 - C) To act as an intermediary between a client and a server, providing filtering and security functions
 - D) To encrypt email messages
- Answer:** C) To act as an intermediary between a client and a server, providing filtering and security functions

27. Which of the following describes the principle of "least privilege" in network security?

- A) Allowing users access to all resources to prevent errors
 - B) Limiting user access to only the resources they need for their role
 - C) Giving administrators full access to all systems at all times
 - D) Ensuring that all users have identical permissions
- Answer:** B) Limiting user access to only the resources they need for their role

28. What is the main purpose of an SSL/TLS certificate?

- A) To prevent network traffic from being intercepted by encrypting it
 - B) To track user activities on the network
 - C) To block unauthorized users from accessing the network
 - D) To provide secure DNS resolution
- Answer:** A) To prevent network traffic from being intercepted by encrypting it

29. What does the concept of "security by design" mean?

- A) Developing a network that is completely secure at all times
- B) Building security measures into the architecture of systems from the outset

- C) Adding security features after deployment
- D) Making security features optional for users

Answer: B) Building security measures into the architecture of systems from the outset

30. What is the primary goal of network monitoring?

- A) To detect security incidents and performance issues in real-time
- B) To provide secure access for remote users
- C) To encrypt all incoming traffic
- D) To provide access to network resources for all users

Answer: A) To detect security incidents and performance issues in real-time

31. What is the role of a digital certificate in security?

- A) To store encryption keys for communication
- B) To verify the authenticity of websites and applications
- C) To compress data for faster transfer
- D) To block malicious websites

Answer: B) To verify the authenticity of websites and applications

32. Which of the following is a key component of a security operations center (SOC)?

- A) Real-time monitoring and response to security incidents
- B) Managing network traffic
- C) Managing user accounts
- D) Running backup operations

Answer: A) Real-time monitoring and response to security incidents

33. What is the primary purpose of network access control (NAC)?

- A) To monitor network performance
- B) To enforce security policies regarding device access to a network
- C) To block malicious network traffic
- D) To encrypt network data

Answer: B) To enforce security policies regarding device access to a network

34. Which security measure helps to protect against SQL injection attacks?

- A) Input validation
- B) Using encryption algorithms
- C) Monitoring network traffic
- D) Applying firewall rules

Answer: A) Input validation

35. What is the main difference between symmetric and asymmetric encryption?

- A) Symmetric encryption uses two keys; asymmetric uses one key
- B) Symmetric encryption is faster but less secure than asymmetric encryption
- C) Symmetric encryption is used for public key exchange
- D) Asymmetric encryption uses the same key for encryption and decryption

Answer: B) Symmetric encryption is faster but less secure than asymmetric encryption

36. Which security practice involves regularly checking for software updates and vulnerabilities?

- A) Patch management
- B) Network segmentation
- C) Incident response
- D) Disaster recovery

Answer: A) Patch management

37. What does a password manager help users to do?

- A) Encrypt passwords for secure storage
- B) Generate and store complex passwords securely
- C) Monitor password usage
- D) Automatically change passwords on a regular basis

Answer: B) Generate and store complex passwords securely

38. What is the purpose of a honeypot in network security?

- A) To attract and trap attackers by simulating vulnerable systems
- B) To encrypt sensitive data
- C) To prevent unauthorized access
- D) To monitor network traffic

Answer: A) To attract and trap attackers by simulating vulnerable systems

39. What type of malware is specifically designed to replicate itself and spread to other devices?

- A) Worm
- B) Trojan horse
- C) Ransomware
- D) Spyware

Answer: A) Worm

40. What is the purpose of a security audit in network security?

- A) To ensure all users have access to the same resources
- B) To examine and assess the effectiveness of security measures and policies
- C) To monitor network traffic for performance issues
- D) To encrypt all data transmissions

Answer: B) To examine and assess the effectiveness of security measures and policies

41. What is the concept of "patching" in the context of network security?

- A) Applying updates to correct vulnerabilities in software
- B) Blocking malicious traffic
- C) Monitoring network activity for suspicious actions
- D) Restricting access to sensitive data

Answer: A) Applying updates to correct vulnerabilities in software

42. Which of the following is NOT a type of network security attack?

- A) Brute force attack
- B) Man-in-the-middle attack
- C) Phishing attack
- D) Data transmission

Answer: D) Data transmission

43. What type of security vulnerability occurs when unauthorized users gain access to a network or system?

- A) Phishing
- B) Data leakage
- C) Unauthorized access
- D) Data integrity breach

Answer: C) Unauthorized access

44. Which type of malware encrypts a victim's files and demands a ransom for decryption?

- A) Virus
 - B) Worm
 - C) Ransomware
 - D) Trojan horse
- Answer:** C) Ransomware

45. What does a security patch typically address?

- A) Vulnerabilities in software or operating systems
 - B) Encryption of data in transit
 - C) Network performance issues
 - D) User authentication problems
- Answer:** A) Vulnerabilities in software or operating systems

46. What is a major risk associated with using weak passwords?

- A) Increased network bandwidth consumption
 - B) Increased likelihood of data breaches or unauthorized access
 - C) Increased encryption requirements
 - D) Decreased system performance
- Answer:** B) Increased likelihood of data breaches or unauthorized access

47. Which of the following is the primary benefit of using two-factor authentication (2FA)?

- A) Faster login process
 - B) Increased security by requiring two forms of identification
 - C) Easier password management
 - D) Reduced need for encryption
- Answer:** B) Increased security by requiring two forms of identification

48. What is the purpose of a network intrusion prevention system (IPS)?

- A) To detect and respond to security threats in real time
 - B) To block all network traffic
 - C) To route data between different networks
 - D) To encrypt all network traffic
- Answer:** A) To detect and respond to security threats in real time

49. What does the term "data integrity" refer to in network security?

- A) Ensuring data is available when needed
 - B) Ensuring data is protected from unauthorized access
 - C) Ensuring data is accurate, consistent, and unaltered
 - D) Ensuring data is transmitted without errors
- Answer:** C) Ensuring data is accurate, consistent, and unaltered

50. What is the primary function of an encryption key in network security?

- A) To verify the authenticity of a user
 - B) To decrypt encrypted data
 - C) To encode and decode data
 - D) To monitor network traffic
- Answer:** C) To encode and decode data

Topic 10 : Basic Working Knowledge of Windows / Linux Operating Systems including Open source operating systems

1. Which of the following is a primary function of an operating system?

- A) Running applications
 - B) Managing hardware resources
 - C) Encrypting data
 - D) Configuring network settings
- Answer:** B) Managing hardware resources

2. What is the default shell used in most Linux distributions?

- A) CMD
 - B) PowerShell
 - C) Bash
 - D) Zsh
- Answer:** C) Bash

3. Which of the following operating systems is an example of an open-source operating system?

- A) Microsoft Windows
 - B) macOS
 - C) Ubuntu
 - D) iOS
- Answer:** C) Ubuntu

4. In Windows, which of the following file systems is commonly used for hard drives?

- A) EXT4
 - B) FAT32
 - C) NTFS
 - D) APFS
- Answer:** C) NTFS

5. Which command is used to list files in a directory in Linux?

- A) dir
 - B) ls
 - C) list
 - D) show
- Answer:** B) ls

6. Which of the following is a command used to change directories in Linux?

- A) move
 - B) cd
 - C) dir
 - D) edit
- Answer:** B) cd

7. What is the default text editor in most Linux distributions?

- A) WordPad
 - B) Notepad
 - C) Vim
 - D) Emacs
- Answer:** C) Vim

8. What is the function of the 'chmod' command in Linux?

- A) To change the file's ownership
- B) To change file permissions
- C) To change the current directory

D) To display the content of a file

Answer: B) To change file permissions

9. In Windows, what is the main function of the Control Panel?

A) To install new software

B) To manage system settings

C) To configure firewall rules

D) To monitor network traffic

Answer: B) To manage system settings

10. Which of the following is a characteristic of open-source operating systems?

A) They are free to use

B) They are typically closed source

C) They require a paid license for updates

D) They are only available for personal use

Answer: A) They are free to use

11. In Linux, which command is used to display the current working directory?

A) pwd

B) dir

C) ls

D) cd

Answer: A) pwd

12. In Windows, which extension is commonly used for executable files?

A) .exe

B) .bat

C) .msi

D) .doc

Answer: A) .exe

13. Which of the following Linux commands is used to install software packages on Ubuntu?

A) yum

B) pacman

C) apt-get

D) dnf

Answer: C) apt-get

14. Which of the following commands in Windows is used to check the system's IP address?

A) ipconfig

B) netstat

C) ping

D) tracert

Answer: A) ipconfig

15. What is the command in Linux to display the system's disk usage?

A) lsblk

B) df

C) du

D) disk

Answer: B) df

16. In Linux, what is the root user's default name?

- A) admin
 - B) superuser
 - C) root
 - D) sys
- Answer:** C) root

17. Which Windows tool is used for disk partitioning and management?

- A) Disk Management
 - B) File Explorer
 - C) Task Manager
 - D) Registry Editor
- Answer:** A) Disk Management

18. Which of the following is the default file system for Linux?

- A) NTFS
 - B) EXT4
 - C) HFS+
 - D) FAT32
- Answer:** B) EXT4

19. In Linux, which directory contains system binary files for essential commands?

- A) /bin
 - B) /usr
 - C) /home
 - D) /etc
- Answer:** A) /bin

20. What is the purpose of the 'sudo' command in Linux?

- A) To execute a command as a normal user
 - B) To change the system settings
 - C) To execute a command with superuser privileges
 - D) To manage user accounts
- Answer:** C) To execute a command with superuser privileges

21. Which of the following file extensions is associated with Linux shell scripts?

- A) .exe
 - B) .sh
 - C) .bat
 - D) .ps1
- Answer:** B) .sh

22. Which of the following is NOT a Windows operating system version?

- A) Windows 10
 - B) Windows XP
 - C) Windows 7
 - D) Ubuntu
- Answer:** D) Ubuntu

23. In Windows, what does the 'Task Manager' allow users to do?

- A) Manage network settings
- B) View and manage running processes
- C) Configure user accounts

D) Install new software

Answer: B) View and manage running processes

24. Which Linux distribution is widely used for servers and workstations?

A) Windows Server

B) Ubuntu

C) Red Hat Enterprise Linux (RHEL)

D) macOS

Answer: C) Red Hat Enterprise Linux (RHEL)

25. In Linux, which file stores user account information?

A) /etc/passwd

B) /etc/shadow

C) /etc/group

D) /home/user

Answer: A) /etc/passwd

26. What is the primary difference between a Windows server and a Windows desktop OS?

A) Windows servers are only for home users

B) Windows servers provide more robust network management and security features

C) Windows desktop OS supports more applications

D) Windows servers cannot run Linux software

Answer: B) Windows servers provide more robust network management and security features

27. Which command is used in Linux to remove a file?

A) del

B) erase

C) rm

D) delete

Answer: C) rm

28. What does the 'ping' command do in both Windows and Linux?

A) Displays the IP configuration

B) Checks the connectivity between two devices

C) Lists all active users on the system

D) Pings the network router

Answer: B) Checks the connectivity between two devices

29. What is a key feature of Linux compared to Windows in terms of software?

A) Linux is proprietary software

B) Linux uses a graphical user interface only

C) Linux is mostly open-source and free

D) Linux is only available for servers

Answer: C) Linux is mostly open-source and free

30. In Windows, which of the following tools is used to check and repair disk errors?

A) Disk Management

B) Check Disk (chkdsk)

C) Disk Cleanup

D) Task Manager

Answer: B) Check Disk (chkdsk)

31. Which of the following commands in Linux is used to display network information?

- A) netstat
 - B) ifconfig
 - C) ipconfig
 - D) traceroute
- Answer:** B) ifconfig

32. Which Linux command allows you to check the current user?

- A) user
 - B) whoami
 - C) id
 - D) current_user
- Answer:** B) whoami

33. Which feature of Linux allows the system to provide access to multiple users at the same time?

- A) File system
 - B) Multitasking
 - C) Multi-user capabilities
 - D) Networking
- Answer:** C) Multi-user capabilities

34. In Windows, which feature allows a user to recover lost or damaged files?

- A) System Restore
 - B) Backup and Restore
 - C) File History
 - D) Disk Management
- Answer:** C) File History

35. Which command is used in Windows to list the contents of a directory?

- A) ls
 - B) dir
 - C) show
 - D) list
- Answer:** B) dir

36. What is the name of the package manager for Red Hat-based Linux distributions?

- A) apt-get
 - B) pacman
 - C) dnf
 - D) yum
- Answer:** D) yum

37. What does the 'ls' command in Linux do?

- A) Changes the directory
 - B) Lists files and directories
 - C) Deletes a file
 - D) Moves a file
- Answer:** B) Lists files and directories

38. What is the function of the 'df' command in Linux?

- A) To delete files
- B) To display disk space usage
- C) To show running processes

D) To list files in a directory

Answer: B) To display disk space usage

39. Which of the following is a Linux-based operating system designed for cloud infrastructure?

A) Windows Server

B) Ubuntu Server

C) CentOS

D) Red Hat Enterprise Linux

Answer: B) Ubuntu Server

40. Which Windows version introduced the 'Start' menu?

A) Windows 95

B) Windows XP

C) Windows 8

D) Windows 10

Answer: A) Windows 95

41. What command is used to display the IP configuration in Windows?

A) netstat

B) ipconfig

C) tracert

D) ping

Answer: B) ipconfig

42. Which file contains configuration information for system settings in Linux?

A) /etc/fstab

B) /etc/passwd

C) /etc/bashrc

D) /etc/hostname

Answer: A) /etc/fstab

43. What is a virtual machine in the context of Linux and Windows?

A) A hardware device that runs multiple OSes

B) A program that runs within an operating system to simulate other hardware

C) A special type of file used by the OS

D) An application that manages system resources

Answer: B) A program that runs within an operating system to simulate other hardware

44. Which of the following is NOT a feature of Linux?

A) Open-source code

B) Multi-user support

C) Multi-tasking capabilities

D) Limited software support

Answer: D) Limited software support

45. Which Windows utility can be used to view and kill processes?

A) Disk Management

B) Task Manager

C) File Explorer

D) Control Panel

Answer: B) Task Manager

46. Which Linux command is used to display system uptime?

- A) uptime
 - B) status
 - C) system
 - D) running
- Answer:** A) uptime

47. In Windows, which of the following is the default directory for user files?

- A) C:\Program Files
 - B) C:\Users
 - C) C:\Windows
 - D) C:\System32
- Answer:** B) C:\Users

48. What does the Linux command 'cat' do?

- A) Changes a file's contents
 - B) Displays the contents of a file
 - C) Removes a file
 - D) Compresses a file
- Answer:** B) Displays the contents of a file

49. In Linux, which file system is commonly used for a root partition?

- A) NTFS
 - B) EXT4
 - C) FAT32
 - D) HFS+
- Answer:** B) EXT4

50. In Windows, which of the following is used to configure system settings and preferences?

- A) Control Panel
 - B) Task Manager
 - C) File Explorer
 - D) Command Prompt
- Answer:** A) Control Panel